Gamma Point Rows under Quantum Wall Crossing and a Criterion for Stable Irrationality

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Abstract

The Gamma integral structure turns the class of a point into a flat covector which reads ordinary rank. We ask when the vanishing or nonvanishing of that covector on a formal-monodromy packet is a birational invariant, and we reduce the question to one marked continuation problem at each threshold of a single equivariant cobordism. Granting that continuation—a gauged-admissible marked Włodarczyk completion, together with marked threshold compatibility (Hypotheses 8.13 and 8.15) on every finite Artin truncation of it—$X \times \mathbf{P}^m$ is irrational for every smooth complex cubic threefold X and every $m \geq 0$. Only the birational maps $X \times \mathbf{P}^m \dashrightarrow \mathbf{P}^{m+3}$ are ever used, so it suffices to assume both inputs for every birational map with these endpoints.

The endpoint contrast is unconditional. Cai's small quantum-connection matrices give a rank-two block with primitive-sixth formal monodromy. We reconstruct its indicial polynomial directly, and an elementary ${}_2F_3$ Barnes calculation shows that the Gamma point row is nonzero on both primitive-sixth lines. Under multiplication by $\mathbf{P}^m$, quantum Künneth produces $m+1$ copies and preserves detection of the resulting packet, whereas projective space itself has no primitive-sixth packet. Detection, rather than multiplicity, is what has to be transported. Through dimension four every blowing-up centre has dimension at most two and carries no primitive-sixth packet of its own, so weak factorization makes the number of copies birationally invariant; a companion paper settles $m = 1$ that way. From $m = 2$ a centre can be a threefold which carries the packet itself, so that vanishing input is unavailable and the packet is marked here by ordinary rank instead.

For the global comparison, two local identities are proved outright: an exact ambient point-column identity at the extremal specialization of a simple VGIT wall, and exact point-row transport across an ordinary flop on a fixed continuation domain. The global argument uses neither. In one equivariant cobordism, the orbit cylinder of a common-open point supplies a single equivariant class whose two Kirwan restrictions are the endpoint point classes, rotation localization kills every intermediate fixed stratum, and Woodward's clutching factorization packages the remaining bubble trees into endpoint tails. A rank-one derived clutching theorem proved here identifies consecutive degrees along each tail, together with their obstruction theories and virtual classes, and makes every tail holonomic and tempered.

The comparison across thresholds is not proved. Beyond gauged-admissibility, the paper assumes an inverse system of one-object marked threshold comparisons, finite at each Artin-energy truncation: at every threshold one marked local Fourier object must induce an isomorphism of the two adjacent cyclic Rees z-modules which intertwines formal monodromy, carries the marked row, and preserves the Stokes and deck data. At a zero mode the comparison is taken on reduced nearby cycles, where it must identify the entire adjacent row-generated module with its reduced realization.

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Given those maps the transport is linear algebra: polynomial functional calculus preserves primary projections, and Rees homogenization makes point-row nonvanishing on a formal-monodromy primary packet birationally invariant. We isolate what would supply those maps: a marked Gamma/window continuation conjecture, new to this paper, implies both hypotheses, and it is the route by which we expect them to be removed. Weaker inputs do not suffice. A two-tail rational counterexample shows that holonomicity of the separate tails does not determine the threshold map, and the incomplete-Gamma and Fourier-boundary countermodels rule out replacements based only on formal monodromy, pairing, integrality, or localized Fourier support.

Keywords. Gamma class; quantum cohomology; birational wall crossing; VGIT; ordinary flop; Stokes matrix; Fourier–Laplace transform; GKZ system; cubic threefold; stable irrationality.

1 Introduction

Let Y be a smooth projective complex variety. Iritani's Gamma integral structure associates a flat section $s_Y(E)$ of the quantum connection to a topological K-class E and identifies the flat pairing with the Euler pairing [Iri23]. If $p \in Y$ is a point, then

$$[s_Y(E), s_Y(\mathcal{O}_p)) = \chi(E, \mathcal{O}_p) = \mathrm{rk}(E). \tag{1}$$

The second entry in (1) is therefore a distinguished flat covector. We call it the *Gamma point row.*

We study whether the zero or nonzero restriction of this row to a formal monodromy packet survives birational modification. A direct composition of one-wall quantum comparisons does not answer the question: adjacent walls use incompatible Novikov completions, and changing a sectorial lift can add a rank-visible Stokes correction. The main construction instead puts the two birational endpoints in one equivariant cobordism and compares their localized gauged theories before either exceptional completion is taken.

The resulting conditional invariant gives a precise route to stable irrationality for every smooth cubic threefold.

Theorem 1.1 (Conditional cubic stabilization criterion)**.** *Let $X \subset \mathbf{P}^4_{\mathbf{C}}$ be a smooth cubic threefold. Assume that every smooth projective birational map admits a Włodarczyk completion which is gauged-admissible in the sense of Definition 8.1 and satisfies the inverse-system family of one-object marked comparisons in Hypotheses 8.13 and 8.15. Then $X \times \mathbf{P}^m$ is irrational for every $m \geq 0$.*

The quantification over all birational maps is for simplicity of statement. The proof uses the two inputs only for birational maps $X \times \mathbf{P}^m \dashrightarrow \mathbf{P}^{m+3}$; the exact weakened form is Remark 8.2.

The $X \times \mathbf{P}^1$ case has a separate dimension-four proof by weak factorization [Rud26]. We keep that argument separate: the present proof introduces a global gauged-localization mechanism which is independent of the stabilization dimension. The case $m = 2$ is the first one not covered by the dimension-four argument.

Why it is not covered is concrete, and it is the reason the present invariant is marked. A blowing-up centre has codimension at least two, so in a fourfold it has dimension at most two; such a centre carries no primitive-sixth packet, which is what makes the multiplicity of the packet birationally invariant through dimension four. In dimension five, centres of dimension three appear. Here X has codimension m in $\mathbf{P}^{m+3}$, so it is itself an admissible centre inside projective space as soon as $m \geq 2$: $\mathrm{Bl}_X\mathbf{P}^5$ is a blowing-up of a rational fivefold along a cubic threefold. Over a common resolution of $X \times \mathbf{P}^m \dashrightarrow \mathbf{P}^{m+3}$, the rational side can therefore acquire packets of the same type as the irrational side, and the vanishing input which made the count birationally invariant is no longer available. Requiring the centre contributions to add rather than to cancel does not restore it: with nonnegative terms a nonzero centre contribution strictly increases the count, and the ascending and descending legs of a factorization can still contribute equally. What is needed instead is a marking for which packets carried by exceptional loci are invisible. The Gamma point row is such a marking, by Euler orthogonality: a class supported on an exceptional locus pairs to zero with the skyscraper of a point of the common birational open. That single orthogonality recurs at three different levels of proof: in the wall functors of Corollary 3.5, which is conditional on its sectorial hypothesis; in the residual summands of the toric calibration of Remark 8.17, which is verified but linear and toric; and in the mutation cone of Conjecture 8.16, which is conjectural.

The conditional conclusion is independent of the classical decomposition of the diagonal and integral Hodge shadows of stable rationality. The companion paper exhibits universally CH_0-trivial cubic threefolds, for which that cycle-theoretic obstruction vanishes, while their first stabilizations remain irrational [Rud26]. The present invariant records irregular quantum-monodromy data instead; it makes no assertion about the Hodge conjecture.

The unconditional endpoint contrast is

$$b(X \times \mathbf{P}^m, \mathcal{P}_6(X \times \mathbf{P}^m)) = 1, \qquad b(\mathbf{P}^{m+3}, \mathcal{P}_6(\mathbf{P}^{m+3})) = 0. \tag{2}$$

where an empty packet has value zero. Here b records whether the point row is nonzero on the indicated packet. These endpoint values use the intrinsic formal-primary projection recovered from the cyclic row; they do not require choosing between alternative sectorial lifts. Section 9 proves (2). Cai's cubic connection supplies the primitive-sixth block; we reconstruct its indicial polynomial from his matrices and compute the point coefficient independently from the cubic hypergeometric period. Thus the cubic-specific source dependency is confined to one endpoint lemma.

The birational statement behind the application is more general.

Theorem 1.2 (Conditional point-row primary invariance)**.** *For smooth projective birational complex varieties whose global cobordism is gauged-admissible and satisfies Hypotheses 8.13 and 8.15, the Gamma point row is nonzero on a formal-monodromy primary packet at one endpoint if and only if it is nonzero on the corresponding packet at the other endpoint.*

The conditional proof uses a smooth projective equivariant completion of one Włodarczyk cobordism. A point in the common birational open sweeps out an orbit cylinder. Its equivariant class restricts to the two endpoint point classes and vanishes on every intermediate fixed stratum, so rotation localization removes every wall contribution.

The rest of the proof has two stages. First, Woodward's factorization packages the complete bubble tree in each neutral affine gauge direction into fixed stable-map factors. Theorem 8.9 identifies their fixed stacks, perfect obstruction theories, virtual classes, and evaluations along each affine tail. The moving factors then give a finite-jet Gamma recurrence, so each tail is holonomic and tempered. Second, Hypotheses 8.13 and 8.15 postulate an inverse-system family of one-object comparisons of the finite cyclic Rees z-modules across the intervening thresholds. Besides preserving formal monodromy and the marked point row, the zero-mode specialization must identify the entire adjacent row-generated cyclic module with its reduced nearby-cycle realization. Primary support is then preserved by polynomial functional calculus. Nonneutral directions are Laurent-finite, and Rees homogenization makes both endpoint maps intertwine formal monodromy with the same half-Tate shift. The resulting cyclic module determines which primary packets the point row detects.

The two standing assumptions have different characters. Gauged-admissibility is a package of finiteness, properness, stability, and marking conditions on one chosen completion. Its numerical-separation clause is expected from equivariant projectivity of the completion; its properness and obstruction-theory clause is the standard shape of the gauged Gromov–Witten package, but no existing theorem supplies it for an arbitrary Włodarczyk completion; its stable-equals-semistable clause is a position assumption on the chamber polarizations. The paragraph following Definition 8.1 records this division clause by clause.

Beyond the gauged-admissibility conditions, the remaining threshold hypothesis continues the marked cyclic module, not the complete nonlinear quantum source. It is finite at each Artin-energy truncation but is a family over all truncations, ordinary degrees, neutral directions, and thresholds. No general geometric threshold is verified in this paper. The comparison is a marked irregular-connection problem and does not follow from the available linear GLSM contour theorem.

The paper also records two local results which motivated the global construction. The first concerns a smooth projective simple VGIT wall $X_- \dashrightarrow X_+$ in the sense of Gu–Yu–Yu, oriented so that $r_+ < r_-$ [GYY25]. Their comparison is a formal, pairing-preserving decomposition

$$\mathrm{QDM}(X_-) \cong \mathrm{QDM}(X_+) \oplus \bigoplus_j \mathrm{QDM}(S)_j. \tag{3}$$

over an exceptional Laurent completion, where S is the wall quotient. Choose corresponding points $p_\pm$ in the common open set.

Theorem 1.3 (Common-open point column)**.** *At the extremal specialization used in the Gu–Yu–Yu comparison, the ambient coordinate of the point class is exact when $r_+ < r_-$:*

$$\mathrm{pr}_{X_+} \Phi([p_-]) = [p_+]. \tag{4}$$

No assertion is made that every wall coordinate of $\Phi([p_-])$ vanishes.

The proof uses the distinguished master-space lift of the common point. Its wall restriction vanishes. Applying the negative-chamber Fourier isomorphism to the lift and to its wall-frequency derivative shows that the derivative is zero in the specialized completed source. Fourier covariance then makes the positive point image horizontal. A regular-singular recursion kills its apparent positive wall-parameter tail.

Our second result is analytic but crepant. Let $Y \dashrightarrow Y'$ be a projective ordinary flop covered by Lee–Lin–Qu–Wang [LLQW16]. Fix one of their analytic-continuation domains in the extremal variable.

Theorem 1.4 (Ordinary-flop point row)**.** *The analytically continued point section of Y equals the intrinsic point section of Y' on the fixed continuation domain. Consequently the point row has the same zero or nonzero restriction to every sectorially realized formal-monodromy packet whose chosen realization is transported by the graph gauge on that branch.*

Pure extremal curves lie in the exceptional locus and cannot meet a point in the common open set. This computes the point column exactly at transverse Novikov degree zero. A divisor pulled back from the common contraction then kills the transverse tail coefficient by coefficient. This is a theorem about one distinguished column; it does not promote the ancestor comparison of [LLQW16] to full descendant invariance.

The discrepant local case contains a further frame comparison. A formal summand, its sectorial lift, and the intrinsic large-radius Gamma section are different objects. Gu–Yu–Yu prove (3) in a Laurent/formal coefficient ring, while Shen–Shoemaker identify Gamma asymptotic classes at an extremal specialization [SS25]. Passing from these results to an intrinsic Gamma point row requires a common pairing-preserving sectorial realization. We state the corresponding one-wall consequence only under that explicit hypothesis.

The negative results show why this distinction is necessary. A rank-two incomplete-Gamma connection has a nonzero Stokes shear from a ζ_6-formal line into a rank-visible ambient line. The example admits a flat integral hyperbolic doubling and remains after tensoring with $\mathbf{Q}[N]/(N^3)$. Localized partial Fourier–Laplace transform kills the constant extension which records the shear. In two Fourier variables, the delta module at the common origin is invisible after either localization but transforms back to a constant full-support module. Boundary support in Fourier coordinates therefore points in the opposite direction from the desired rank conclusion.

At the local level, the surviving two-wall statement is one-sided. Complete ray-ordered corrections must have rank-zero *targets*; their sources and scalar coefficients are irrelevant. Under that hypothesis, a finite factorization preserves the point row by elementary linear algebra. A signed punctual coefficient of an oriented boundary complex is a plausible shadow of these factorwise conditions, but it is not equivalent to them in the local formalism. A single alternating coefficient can vanish by cancellation. Under Hypotheses 8.13 and 8.15, the global construction in Section 8 bypasses this missing marked comparison.

Sections 2–4 establish the point-row formalism and the local identities. Sections 5 and 6 give the countermodels, and Section 7 isolates the local composition condition they force. Section 8 proves the Gamma-ratio reduction and conditional Theorem 1.2 by the common-master method. Section 9 proves (2) and Theorem 1.1. Section 10 records the source, analytic, and verification boundaries.

2 The point row and three distinct frames

We fix the pairing convention before discussing wall crossing. Let $\mathcal{S}(Y)$ denote a space of flat sections of the quantum connection on a domain where the Gamma framing is defined. Iritani's section associated with $E \in K^0_{\mathrm{top}}(Y)$ has the form

$$s_Y(E) = L_Y(\tau, z) z^{-\mu} z^{c_1(Y)} (2\pi)^{-\dim Y/2} \widehat{\Gamma}_Y (2\pi i)^{\deg/2} \operatorname{ch}(E), \tag{5}$$

up to the standard choice of normalization. The flat pairing is

$$[s_1, s_2) := (s_1(\tau, e^{-\pi i} z), s_2(\tau, z))_Y, \qquad [s_Y(E), s_Y(F)) = \chi(E, F). \tag{6}$$

See [Iri23, Section 1].

Definition 2.1 (Gamma point row)**.** For $p \in Y$, the Gamma point row is

$$\mathfrak{r}_{Y,p}(v) := [v, s_Y(\mathcal{O}_p)). \tag{7}$$

On Gamma-framed K-classes it is ordinary rank:

$$\mathfrak{r}_{Y,p}(s_Y(E)) = \chi(E, \mathcal{O}_p) = \operatorname{rk}(E). \tag{8}$$

Putting the point in the first entry instead would introduce the usual dimension sign. We use (7) throughout.

Three levels of structure occur in the arguments below.

(i) A *formal packet* is a generalized formal-monodromy or exponential block of the $z = 0$ formal connection.

(ii) A *sectorial lift* is a choice of actual flat subspace with that formal asymptotic expansion on an admissible sector.

(iii) The *intrinsic point section* is the Gamma-normalized flat section fixed by the large-radius fundamental solution.

A positive-z gauge can identify formal quantum D-modules without identifying (iii) across two incompatible Novikov completions. Likewise, two sectorial lifts of the same formal packet can differ by a Stokes shear. The zero or nonzero restriction of (7) is therefore not determined by the formal packet alone.

We write $\mathcal{P}_6(Y)$ for the generalized formal-monodromy block with eigenvalue $\zeta_6 = e^{2\pi i/6}$. When $\mathcal{P}_6(Y)$ appears inside a point-row Boolean, it denotes a specified sectorial lift of that block; a different lift is different data. An empty block has Boolean value zero.

Definition 2.2 (Point-row Boolean)**.** For a sectorially realized formal packet $\mathcal{P} \subset \mathcal{S}(Y)$, put

$$b(Y, \mathcal{P}) = \begin{cases} 1, & \mathfrak{r}_{Y,p}|_{\mathcal{P}} \neq 0, \\ 0, & \mathfrak{r}_{Y,p}|_{\mathcal{P}} = 0. \end{cases} \tag{9}$$

The paper's exact wall identities either identify the intrinsic point section directly or remain within one specified receiver. Whenever an additional sectorial realization is required, it is stated as a hypothesis.

3 The exact point column at a simple VGIT wall

Let $X_- \dashrightarrow X_+$ be a simple VGIT wall crossing satisfying the hypotheses of [GYY25, Definition 3.2]: the chamber quotients and wall quotient S are smooth and projective, and the normal weights at the wall are ± 1. Assume $r_+ < r_-$ and write $\nu = r_- - r_+$.

Proposition 3.1 (Pinned simple-wall input)**.** *Fix Gu–Yu–Yu, arXiv:2508.15770v1, and a simple wall as above. The following are the only inputs from that source used in this section.*

(G1) *Their decomposition theorem gives a pairing-preserving, connection-intertwining decomposition over the exceptional Laurent completion [GYY25, Theorem 6.2, p. 54]:*

$$\Phi : \mathrm{QDM}(X_-)^{\mathrm{La,red}} \longrightarrow (\tau_+^{\mathrm{red}})^* \, \mathrm{QDM}(X_+)^{\mathrm{La,red}} \oplus \bigoplus_{j=0}^{\nu-1} (\zeta_j^{\mathrm{red}})^* \, \mathrm{QDM}(S)^{\mathrm{La,red}}. \tag{10}$$

(G2) *For common-open points $p_\pm$, their Kirwan lift construction gives a class a_p with [GYY25, Lemma 3.27, pp. 24–25]*

$$\kappa_+(a_p) = [p_+], \qquad \kappa_-(a_p) = [p_-], \qquad a_p|_{F_0} = 0. \tag{11}$$

Here $\kappa_\pm : H^*_{\mathbf{C}^*}(W) \to H^*(X_\pm)$ are the ordinary cohomological Kirwan maps. The final equality is the point-support specialization proved in that lemma: if $[p_+]|_{\mathbf{P}(N_{F_0,+})} = f(h_+)$, then its proof identifies $a_p|_{F_0} = f(\lambda)$. A common-open point misses the positive exceptional locus, so $f = 0$.

(G3) *At the extremal specialization,*

$$\mathrm{FT}_{X_+}(\alpha) = \kappa_+(\alpha) + \sum_{k>0} f_{+,k}(\alpha) q_{\mathrm{w}}^k, \qquad f_{+,k}(\alpha) \in H^*(X_+)[z].$$

On the negative side, if $\deg_\lambda(\alpha|_{F_0}) \leq r_{F_0,-} - 1$*, then* $\mathrm{FT}_{X_-}(\alpha) = \kappa_-(\alpha)$ *exactly at the same specialization. The completed master source is finite free, and the negative Fourier map is an isomorphism after this base change [GYY25, Proposition 5.2, Lemmas 5.8 and 5.10, Proposition 5.9 and Theorem 5.5, pp. 37, 42–44].*

(G4) *Fourier transformation is covariant for the wall-frequency connection: it carries multiplication by the equivariant wall class to the receiver's wall logarithmic derivative together with quantum multiplication by its Kirwan image and the receiver-coordinate derivative [GYY25, Proposition 4.14(2), p. 32; Proposition 4.21, p. 36].*

No sectorial or Gamma comparison is included in this input.

Proof. The ring in (G1) is Laurent in a fractional wall variable and complete in the remaining Novikov and bulk variables. The four clauses are the cited decomposition, Kirwan-lift, leading-term, and covariance statements, respectively; no additional conclusion is drawn from them here. We use only the extremal specialization below. □

Let $p_\pm$ be corresponding points in the common open set. Gu–Yu–Yu's lift a_p is the one in Proposition 3.1(G2).

Theorem 3.2 (Exact ambient point coordinate)**.** *After setting the master Novikov and bulk variables to zero while retaining the wall variable, the ambient coordinate of* (10) *satisfies*

$$\mathrm{pr}_{X_+} \Phi([p_-]) = [p_+]. \tag{12}$$

This theorem makes no assertion about the centre coordinates of $\Phi([p_-])$*.*

Proof. Let $q_{\mathrm{w}} = S_{F_0}$ denote the wall variable, let λ be the equivariant wall frequency, and put $D = \kappa_+(\lambda)$. Proposition 3.1(G3) applies to a_p because its wall restriction is zero. At the specialization under consideration, it gives

$$\mathrm{FT}_{X_-}(a_p) = [p_-]. \tag{13}$$

Clause (G3) gives only

$$\mathrm{FT}_{X_+}(a_p) = [p_+] + O(q_{\mathrm{w}}). \tag{14}$$

Apply (G3) again to λa_p. Its wall restriction is zero and

$$\kappa_-(\lambda a_p) = \kappa_-(\lambda)[p_-] = 0. \tag{15}$$

by degree. The finite-free/isomorphism clause of (G3) survives base change to the extremal specialization. Equation (15) and (G3) therefore imply that λa_p is zero in the specialized completed source. In particular,

$$\mathrm{FT}_{X_+}(\lambda a_p) = 0. \tag{16}$$

The Fourier covariance of Proposition 3.1(G4) rewrites (16) as

$$\left(zq_{\mathrm{w}}\partial_{q_{\mathrm{w}}} + D \star_{\tau_+(q_{\mathrm{w}})} + (q_{\mathrm{w}}\partial_{q_{\mathrm{w}}}\tau_+(q_{\mathrm{w}}))\star_{\tau_+(q_{\mathrm{w}})}\right) \mathrm{FT}_{X_+}(a_p) = 0. \tag{17}$$

Here $\tau_+(q_{\mathrm{w}})$ is the specialized receiver coordinate. Write

$$\mathrm{FT}_{X_+}(a_p) = [p_+] + \sum_{k>0} q_{\mathrm{w}}^k v_k, \qquad \tau_+(q_{\mathrm{w}}) = f(q_{\mathrm{w}})\mathbf{1} + \eta(q_{\mathrm{w}}), \tag{18}$$

where $f(0) = 0 = \eta(0)$ and $\eta(q_{\mathrm{w}})$ has positive cohomological degree. Clause (G3) gives $v_k \in H^*(X_+)[z]$. Every nonconstant stable map in a positive multiple of the wall fibre is contained in the exceptional locus and misses p_+. Thus the positive-wall quantum corrections and $\eta(q_{\mathrm{w}})$ annihilate $[p_+]$.

We prove simultaneously that every v_k and every coefficient of f vanishes. Suppose this is known below order $m > 0$, and write $[q_{\mathrm{w}}^m]f = c_m$. The coefficient of q_{w}^m in (17) is

$$(zm\ \mathrm{id} + D \cup -)v_m + mc_m[p_+] = 0. \tag{19}$$

If $v_m \neq 0$, the highest power of z in zmv_m cannot be cancelled by either $D \cup v_m$ or the constant term $mc_m[p_+]$, because v_m is polynomial in z. Hence $v_m = 0$, and then $c_m = 0$. Induction kills the complete tail and the possible unit part of the receiver coordinate, proving (12). □

Warning 3.3. The continuous Fourier map applies stationary phase to the transformed master fundamental solution. Positive master-Novikov terms can survive even though the classical restriction $a_p|_{F_0}$ is zero. The argument proves the ambient coordinate exactly because the wall-frequency derivative dies in the specialized completed source; polynomiality from (G3) then controls the possible receiver unit correction.

The following consequence isolates the extra analytic premise required to turn (12) into a Gamma-row statement.

Hypothesis 3.4 (One-wall sectorial realization)**.** At a fixed nonzero wall parameter there is a pairing-preserving sectorial realization of (10) on one common admissible sector such that:

(a) the ambient formal summand realizes the intrinsic sectorial solution space of X_+;

(b) the wall summands realize the Gamma spans of the wall functors in the Shen–Shoemaker semiorthogonal decomposition;

(c) the realized point row has ambient coordinate (12).

Corollary 3.5 (Conditional wall-local rank identity)**.** *Under Hypothesis 3.4, the point row restricts to the point row of X_+ on the ambient packet and vanishes on every wall packet. In particular, for the whole generalized ζ_6 packet in this one receiver,*

$$b(X_-, \mathcal{P}_6(X_-)) = b(X_+, \mathcal{P}_6(X_+)). \tag{20}$$

Proof. The ambient assertion is Theorem 3.2 together with the pairing-preserving realization. A wall functor has image supported on the exceptional locus. Pairing its Gamma class with the point class in the common open gives zero Euler characteristic. Shen–Shoemaker's oriented Gamma asymptotics identify these supported classes with the wall blocks on the chosen common sector [SS25]. Hence the point row vanishes there. Restricting the resulting whole-block identity to the generalized ζ_6 packet gives (20). □

The hypothesis is not a reformulation of the cited formal theorems. Gu–Yu–Yu work over an exceptional Laurent completion; Shen–Shoemaker's asymptotic theorem is extremal and sectorial. Identifying the intrinsic large-radius point section with the full ambient packet across those frames is additional data.

4 A path-local theorem for ordinary flops

Let $Y \dashrightarrow Y'$ be a projective ordinary flop covered by [LLQW16], and let ℓ, ℓ' be its effective extremal fibre classes. The graph correspondence F identifies the big quantum products and Poincaré pairings after analytic continuation

$$q' = q^{-1}, \qquad F(\ell) = -\ell'. \tag{21}$$

The continuation is analytic in the extremal variable and formal in the other Novikov variables.

Fix a simply connected nonsingular domain U for the continued extremal variable. We use the hybrid coefficient ring

$$\mathcal{O}(U)\widehat{\otimes}\Lambda_{\perp}[[\tau]]. \tag{22}$$

where $\Lambda_{\perp}$ is the transverse Novikov completion and τ denotes the bulk variables; the coefficient recursion below is taken after extension to $\mathbf{C}((z))$. The *graph gauge* is the horizontal gauge obtained from the Lee–Lin–Qu–Wang comparison of big quantum connections on U, normalized by the classical graph correspondence. Thus it acts on flat columns, not only on quantum products.

Choose corresponding points $p \in Y$ and $p' \in Y'$ outside the exceptional loci. Let A be an ample divisor on the common contraction and write D, D' for its pullbacks. Then $D \cdot \ell = D' \cdot \ell' = 0$, while D is positive on every effective class away from the extremal ray. We complete the transverse Novikov ring by D-degree.

Theorem 4.1 (Point-section preservation)**.** *On U, the graph gauge satisfies*

$$F(s_Y(\mathcal{O}_p)) = s_{Y'}(\mathcal{O}_{p'}). \tag{23}$$

Consequently it preserves the Gamma point row.

Proof. At transverse degree zero, every nonconstant stable map has pure extremal class and lies in the exceptional locus. It cannot meet the chosen point. All positive-extremal descendant invariants with the point insertion therefore vanish. The point columns on both sides are their common classical column for every nonsingular value of q: $\widehat{\Gamma}_Y \operatorname{ch}(\mathcal{O}_p)$ is a scalar multiple of the top-degree class $[p]$, on which higher Gamma and Chern-class terms vanish. Analytic continuation of this identically classical extremal column remains classical, and the graph correspondence sends $[p]$ to $[p']$.

Let

$$\delta = F(s_Y(\mathcal{O}_p)) - s_{Y'}(\mathcal{O}_{p'}).$$

It is jointly flat, meaning flat for both the z- and Novikov-direction connections on the hybrid ring (22), and has zero transverse constant term. Suppose δ_β is a nonzero coefficient of minimal positive D-degree. Take coefficients modulo the extremal ray. By minimality, every positive-transverse convolution term involves a lower D-degree coefficient and vanishes. Horizontality for the Novikov derivation defined by D therefore gives

$$\big((D \cdot \beta)\operatorname{id} + z^{-1}(D \star_q -)\big)\delta_\beta = 0. \tag{24}$$

At transverse degree zero, quantum corrections to $D\star_q$ have pure extremal degree. The divisor axiom multiplies them by $D \cdot \ell = 0$. Thus $D\star_q = D \cup -$ in (24). Cup product by D is nilpotent and $D \cdot \beta > 0$, so the displayed operator is invertible over $\mathbf{C}((z))$. This contradiction kills the first possible coefficient. Induction proves (23). Flatness of the pairing then preserves the row (7). □

Corollary 4.2 (Packetwise Boolean)**.** *Let $\mathcal{P} \subset \mathcal{S}(Y)$ be a sectorially realized formal-monodromy packet whose chosen realization is transported by the Lee–Lin–Qu–Wang graph gauge to a sectorially realized packet $\mathcal{P}' \subset \mathcal{S}(Y')$. On the fixed continuation domain,*

$$b(Y, \mathcal{P}) = b(Y', \mathcal{P}'). \tag{25}$$

Proof. The graph gauge preserves the quantum connection, grading, first Chern class, and Poincaré pairing. It therefore transports the complete $z = 0$ formal type. Theorem 4.1 identifies the covector on the transported sectorial realization, and (25) follows. □

The theorem is narrower than a Gamma/Fourier–Mukai comparison for every K-class. Pure extremal curves need not miss the support of a general class, and the support computation in the first paragraph of the proof would fail. Nor does the argument assert phase independence outside the chosen continuation domain.

5 A minimal ambient-target Stokes shear

Set $\alpha = 1/6$ and consider the rank-two meromorphic connection

$$\frac{d}{dz}\begin{pmatrix} y_0 \\ y_1 \end{pmatrix} = \begin{pmatrix} 0 & z^{-2} \\ 0 & z^{-2} + \alpha z^{-1} \end{pmatrix} \begin{pmatrix} y_0 \\ y_1 \end{pmatrix}. \tag{26}$$

Proposition 5.1 (Incomplete-Gamma countermodel)**.** *Connection* (26) *has a formal line of monodromy ζ_6 whose sectorial lift undergoes a nonzero Stokes shear into a monodromy-one line. The shear can be equipped with a flat hyperbolic pairing preserving a $\mathbf{Z}[\zeta_6]$-lattice and can be tensored with $\mathbf{Q}[N]/(N^3)$ without disappearing. Hence formal monodromy, pairing, integrality, and a commuting length-three nilpotent tag do not determine the zero or nonzero restriction of a flat covector to the ζ_6 line.*

Proof. Two flat columns are

$$f_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad f_\alpha(z) = \begin{pmatrix} \Gamma(1-\alpha, 1/z) \\ z^\alpha e^{-1/z} \end{pmatrix}. \tag{27}$$

Indeed,

$$\frac{d}{dz}\Gamma(1-\alpha, 1/z) = z^{\alpha-2} e^{-1/z}, \tag{28}$$

and direct differentiation verifies both rows of (26). The formal factors are the trivial line and $e^{-1/z} z^\alpha$, whose formal monodromy is $e^{2\pi i \alpha} = \zeta_6$.

Analytic continuation of the incomplete Gamma function adds a nonzero multiple of the complete Gamma value. After rescaling f_α, the Stokes jump has the form

$$f_\alpha^+ = f_\alpha^- + f_0. \tag{29}$$

Choose a flat covector r with $r(f_0) = 1$ and $r(f_\alpha^-) = 0$. Then $r(f_\alpha^+) = 1$. Thus the Boolean restriction of r to the sectorial lift of the same formal line changes.

Let $A(z)$ be the matrix in (26) and double the system by its dual:

$$B(z) = \operatorname{diag}(A(z), -A(z)^t), \qquad J = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}. \tag{30}$$

Then $B^t J + JB = 0$. If the jump on the first summand is $S = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, the doubled jump $\operatorname{diag}(S, S^{-t})$ preserves J and the obvious $\mathbf{Z}[\zeta_6]$-lattice. Finally tensor with $\mathbf{Q}[N]/(N^3)$ and let the jump act trivially on this factor. It commutes with N and still has (29). This proves the assertion. □

The example also occurs in a flat confluence family. Replacing z by z/t gives the z-equation

$$\frac{dY}{dz} = \begin{pmatrix} 0 & tz^{-2} \\ 0 & tz^{-2} + \alpha z^{-1} \end{pmatrix} Y, \tag{31}$$

with compatible t-equation inherited from the pullback connection. The two exponential factors collide at $t = 0$, while the nonzero Stokes coefficient persists on the ramified punctured t-line. Formal information at the confluent fibre cannot recover its target orientation.

Remark 5.2. Proposition 5.1 is an analytic no-go theorem, not a geometric counterexample. We do not assert that (26) is the quantum connection of a smooth projective wall crossing. It proves only that a geometric proof must use information absent from the listed formal shadows.

6 What localized Fourier–Laplace forgets

The incomplete-Gamma jump has an elementary differential-algebraic shadow. For $g(x) = \Gamma(1-\alpha, x)$,

$$(\partial_x + 1 + \alpha/x)\partial_x g = 0. \tag{32}$$

Differentiation forgets the constant solution, and that constant is the additive datum in the continuation formula.

Let $D_t = \mathbf{C}\langle t, \partial_t \rangle$. Under algebraic Fourier transform,

$$\mathbf{C}[t] = D_t/D_t\partial_t \quad \longmapsto \quad \delta_0 = D_\tau/D_\tau\tau. \tag{33}$$

Localizing at τ kills δ_0. Localized partial Fourier–Laplace therefore cannot recover the extension constant in (32). The four-term GKZ/Gauss–Manin comparison in [RSSW21] uses the same nonconservativity.

Proposition 6.1 (Punctual Fourier corner)**.** *Let*

$$\delta_{00} = D_{\tau_1,\tau_2}/D_{\tau_1,\tau_2}(\tau_1, \tau_2).$$

Localization at either τ_i kills δ_{00}, whereas

$$\mathrm{FL}^{-1}(\delta_{00}) = \mathcal{O}_{\mathbf{A}^2} \tag{34}$$

up to shift and sign conventions. Thus an object supported at the intersection of two deleted Fourier boundaries can have generic-rank-one inverse transform.

Proof. Use the Weyl-algebra Fourier automorphism $x_i \mapsto -\partial_{\tau_i}$ and $\partial_{x_i} \mapsto \tau_i$. It exchanges $D_x/D_x(\partial_{x_1}, \partial_{x_2})$ with δ_{00}. Since τ_i acts locally nilpotently on the latter, inverting either coordinate kills it. The inverse transform is the constant module, which has generic rank one. $\square$

Tensoring Proposition 6.1 with the ζ_6 rank-one connection, its hyperbolic dual, and $\mathbf{Q}[N]/(N^3)$ preserves every conclusion. Bare double-boundary support is therefore not a rank-zero condition.

Nor does marking one punctual vector make it unique. Let C be any finite-dimensional coefficient space and put $\mathcal{B} = \delta_{00} \otimes C$. For $G \in \mathrm{GL}(C)$, two receivers can differ by G while having the same support and characteristic cycle. After hyperbolic doubling, $G \oplus G^{-t}$ preserves a perfect pairing. It can still change a chosen point row. Consequently, a claim that the punctual quotient is generated by one marked line is additional geometry.

Enlarging the discarded subspace to the kernel of the point row makes the quotient one-dimensional only tautologically: invariance of that kernel is the theorem one seeks.

The unlocalized boundary map

$$\mathrm{FL}(M) \longrightarrow j_*j^*\,\mathrm{FL}(M), \qquad j : \mathbf{G}_m \hookrightarrow \mathbf{A}^1, \tag{35}$$

retains the missing object. In a two-coordinate problem one must also retain the orientation, duality, and $! \to *$ or can / var maps of the complete Cech boundary square. Proposition 6.1 shows that retaining the object is necessary, not that its support alone proves the desired vanishing.

7 The two-wall criterion

Let Y be an intermediate smooth projective variety, $D \subset Y$ the union of the two incident wall boundaries, and $p \in U := Y \setminus D$. Suppose a common analytic receiver contains the two sectorial realizations under comparison, and let T be their transition. Write $\mathcal{S}_D(Y)$ for a span of Gamma sections represented by classes supported on D. The point row annihilates this span:

$$\mathfrak{r}_{Y,p}(v) = 0 \qquad (v \in \mathcal{S}_D(Y)). \tag{36}$$

Hypothesis 7.1 (Rank-zero-target condition)**.** On the packet under consideration, the relative transition admits a canonical ray-ordered factorization

$$T = \prod_{a=1}^{m} (1 + v_a \otimes \lambda_a) \tag{37}$$

whose complete residue targets satisfy

$$\mathfrak{r}_{Y,p}(v_a) = 0 \qquad (1 \leq a \leq m). \tag{38}$$

The word *complete* is essential. Refining a residue block into coordinate terms can create nonzero ranks which cancel only after the full block is assembled. The boundary-support condition $v_a \in \mathcal{S}_D(Y)$ is a geometric sufficient condition for (38), by (36), but it is not part of the abstract hypothesis.

Theorem 7.2 (One-row assembly)**.** *Under Hypothesis 7.1,*

$$\mathfrak{r}_{Y,p} T = \mathfrak{r}_{Y,p}. \tag{39}$$

Consequently, any finite factorization whose ordinary-flop steps satisfy Theorem 4.1 and whose discrepant adjacent-wall transitions satisfy Hypothesis 7.1 preserves the point-row Boolean on the transported packet.

Proof. Equation (38) gives

$$\mathfrak{r}_{Y,p}(1 + v_a \otimes \lambda_a) = \mathfrak{r}_{Y,p}$$

for every factor in (37). Multiplying the identities proves (39). The factorization statement follows by induction, using Theorem 4.1 at crepant steps. □

A stronger geometric condition is one-sided kernel support. If the cone of the transition kernel relative to the diagonal has support

$$\mathrm{Supp}(K_{\mathrm{rel}}) \subset Y \times D, \tag{40}$$

with D in the output factor, and if its induced action satisfies $\operatorname{im}(T - \mathrm{id}) \subset \mathcal{S}_D(Y)$, then (36) gives $\mathfrak{r}_{Y,p}T = \mathfrak{r}_{Y,p}$ directly. This conclusion does not require, and does not imply, that every factor in an independently chosen factorization of T has supported target. Support in $D \times Y$ instead constrains the source and would not imply (39).

There is a tempting singular shadow of the factorwise condition. Let B_{corner} be a complete oriented correction left by the two localizations after subtracting the common point contribution, and let its signed punctual multiplicity be the alternating δ_{00} coefficient

$$\mu_{00}(B_{\mathrm{corner}}).$$

Fourier transform exchanges a punctual coefficient with a zero-section multiplicity, hence with generic output rank in the elementary model of Proposition 6.1. One might therefore hope for

$$\mu_{00}(B_{\mathrm{corner}}) = 0 \tag{41}$$

as a numerical test. This implication is *not* proved here. A single alternating multiplicity can vanish by cancellation even when individual ray targets have nonzero point row, and no exact point-row marking on B_{corner} has yet been constructed. Proposition 6.1 also explains why merely knowing that the boundary object is punctual proves nothing.

The failure of a global-to-factorwise inference already occurs in dimension two. Let $r = e_1^*$ on $V = \mathbf{C}^2$, and put $A = e_1 \otimes e_2^*$. Then $A^2 = 0$ and

$$\mathrm{id} = (\mathrm{id} + A)(\mathrm{id} - A), \tag{42}$$

although both elementary factors have target e_1 and $r(e_1) = 1$. Thus even a vanishing total correction does not force the targets in a chosen ray factorization to lie in $\ker r$.

The missing comparison can be stated blockwise: for each complete ray block one would need an unlocalized boundary object B_a and an identity

$$\mathfrak{r}_{Y,p}(v_a) = \mu_{00}(B_a). \tag{43}$$

with enough concentration or effectivity to prevent cancellation inside the block. Equation (43) and $\mu_{00}(B_a) = 0$ for every a would imply Hypothesis 7.1; the single global equation (41) does not.

Remark 7.3 (What remains to be proved)*.* Neither (37), the action condition following (40), nor the blockwise comparison (43) is deduced here for a general pair of discrepant walls. Existing one-wall theorems determine formal decompositions and, in special toric settings, Gamma/window comparisons. The missing statement is directed and two-wall: it must retain which side of every vanishing cycle is the output and must compare that orientation with the Gamma point row.

8 A common master and birational transport

The obstruction in Sections 5–7 comes from comparing two local receivers at an intermediate chamber. We now avoid that comparison. A single equivariant cobordism carries both endpoint quantum Kirwan maps, and virtual localization compares their point rows before either endpoint Novikov completion is taken.

8.1 The common point row

Definition 8.1 (Admissible marked completion)**.** A smooth projective $\mathbf{G}_m$-completion W, two extreme linearizations, and a point p in their common birational open are *gauged-admissible* if the following conditions hold.

(i) The extreme quotients Y_- and Y_+ are smooth and free, and stable equals semistable for the chamber polarizations used to define their gauged theories. This equality is not imposed at a critical wall polarization; the wall is handled by the master stacks in (ii).

(ii) Degree by degree, the polarization master stacks are proper Deligne–Mumford stacks with Woodward's relative perfect obstruction theories, and the localized large-area formula applies coefficientwise.

(iii) Finitely many equivariant numerical divisor classes separate every allowed total equivariant degree, and the effective infinity-side degree semigroup is pointed.

(iv) There is a class $a_p \in H^*_{\mathbf{G}_m}(W)$ whose ordinary Kirwan restrictions are the two endpoint point classes and whose restriction to every intermediate fixed component is zero.

These are assumptions on the gauged completion, not consequences of smooth projectivity alone.

The four conditions have different standing. Condition (iii) is closest to automatic: equivariant projectivity of the completion is expected to supply the separating divisor classes, and pointedness of the infinity-side degree semigroup is a positivity property of the chosen completion. Condition (ii) has the shape of the properness, obstruction-theory, and large-area localization package developed by Woodward for gauged maps [Woo18]; what is assumed is that this package applies to the polarization master stacks of the given completion at every degree used, and no existing theorem asserts this for an arbitrary Włodarczyk completion. Condition (i) is a position assumption on the chamber polarizations: stable equals semistable is the standard VGIT genericity condition, and we do not prove that every completion admits polarizations satisfying it. The class in condition (iv) is constructed below by the orbit-cylinder argument following Proposition 8.4.

Remark 8.2 (Endpoint-only hypotheses for the cubic application). Theorem 1.1 needs gauged-admissibility and the threshold comparisons of Hypotheses 8.13 and 8.15 only for the birational maps $X \times \mathbf{P}^m \dashrightarrow \mathbf{P}^{m+3}$. Its proof applies Theorem 8.20 once for each m, to a map supplied by the assumed rationality. It therefore suffices to assume: for every $m \geq 0$ and every birational map $\varphi\colon X \times \mathbf{P}^m \dashrightarrow \mathbf{P}^{m+3}$, some Włodarczyk completion of φ is gauged-admissible and satisfies both threshold hypotheses. The universal quantifier over maps with these endpoints cannot be dropped, because the rationality being contradicted supplies an unknown map. The weakened assumption is still an infinite family, over all m, all such maps, and all Artin levels, ordinary degrees, neutral directions, and thresholds of the chosen completions, but it no longer quantifies over arbitrary pairs of birational varieties.

Fix such a completion. We work coefficientwise at sufficiently large area. Write $\kappa_\pm$ for the corresponding quantum Kirwan maps, write $\mathrm{Kir}_\pm : H^*_{\mathbf{G}_m}(W) \to H^*(Y_\pm)$ for the ordinary cohomological Kirwan maps, and write $\tau_{Y_\pm,-}$ for Woodward's localized graph potentials. The derivatives

$$A_\pm = D\tau_{Y_\pm,-}\, D\kappa_\pm. \tag{44}$$

are the endpoint localized gauged maps; this is the localized adiabatic identity [Woo18, Equation (68)].

The marked class has the restrictions

$$\mathrm{Kir}_\pm(a_p) = [p_\pm], \qquad a_p|_F = 0. \tag{45}$$

for every fixed component F occurring at an intermediate polarization. Here p_- and p_+ lie in the common birational open set.

Lemma 8.3 (Point insertion and the Gamma row). *Let R be a finite Artin quotient of the Novikov–bulk coefficient ring. Write $\mathcal{H}_Y(R)$ for the cohomological input module and $\mathcal{S}_Y(R)$ for the corresponding module of flat sections. With Woodward's graph-space normalization,*

$$D\tau_{Y,-} : \mathcal{H}_Y(R) \longrightarrow \mathcal{S}_Y(R).$$

If $\epsilon_{Y,p}$ is the normalized Poincaré covector defined by a point insertion and $n = \dim Y$, then there is an invertible scalar $c_n(z)$, depending only on n and the two normalization conventions, such that

$$\mathfrak{r}_{Y,p}\, D\tau_{Y,-} = c_n(z)\epsilon_{Y,p}. \tag{46}$$

Consequently the endpoint functional with distinguished insertion $\mathrm{Kir}(a_p) = [p]$ is

$$c_n(z)^{-1}\mathfrak{r}_{Y,p} D\tau_{Y,-} D\kappa.$$

Proof. The Gamma framing of $\mathcal{O}_p$ is the graph fundamental solution applied to the top cohomology class. The Gamma and Chern-class factors act trivially on that class. The remaining dimension sign, power of z, and power of $2\pi i$ are invertible and depend only on n and on the chosen Gamma and graph-space normalizations; their product is $c_n(z)$. Unitarity of the fundamental solution then identifies pairing against $s_Y(\mathcal{O}_p)$ with $c_n(z)$ times the Poincaré point covector. This is (46), and composition with $D\kappa$ gives the last assertion. The argument is coefficientwise over R, so it commutes with input and bulk differentiation. $\square$

Proposition 8.4 (Support collapse). *At every fixed equivariant degree, the point-marked virtual Kalkman formula has no intermediate-polarization contribution. Hence its two endpoint coefficients agree in the extended Novikov coefficient-function space. After Rees homogenization and in any common realization of the two endpoint series, this reads*

$$\mathfrak{r}_{Y_-,p_-} A_- = \mathfrak{r}_{Y_+,p_+} A_+. \tag{47}$$

The identity is coefficientwise in all remaining Novikov and bulk variables; the proposition does not construct the common realization.

Proof. Retain rotation of the parametrized graph curve, with equivariant parameter $\hbar$. At each ordinary input marking insert the normalized equivariant point class of $0 \in \mathbf{P}^1$. Rotation localization then puts every input on the zero-side bubble tree. Next choose equivariant divisor classes $D_1, \ldots, D_s$ spanning the full equivariant numerical dual of the allowed total degrees; their endpoint Kirwan restrictions in particular span the endpoint numerical divisor spaces. Insert Woodward's Liouville classes with parameters t_j. On a graph fixed locus their contribution is

$$\exp\left(\sum_j t_j D_j\right) \prod_j x_j^{D_j \cdot \widetilde{d}_+}, \qquad x_j = \exp(t_j \hbar). \tag{48}$$

where $\widetilde{d}_+$ is the total infinity-side equivariant degree: the bubble degree together with the affine cocharacter degree of the gauged principal component [Woo18, Corollary 9.10(c)]. Extracting the trivial character in all x_j forces $\widetilde{d}_+ = 0$. The infinity factor is therefore the unstable identity: positivity on the effective infinity-side semigroup leaves no nonconstant degree-zero component. The zero factor is exactly the localized graph potential in (44).

Using the full equivariant divisor lattice is important: endpoint divisor spaces alone need not separate an affine cocharacter degree.

For each fixed degree and sufficiently large area, every nonendpoint polarization-fixed graph has its principal component in a fixed quotient. Moreover every marking and node lands in the corresponding fixed locus [GW15, Proposition 3.15(c), Lemma 3.17 and Proposition 3.18]. Its insertion therefore factors through one of the restrictions in (45) and vanishes before division by the virtual normal Euler class. Only the two endpoints survive the virtual Kalkman identity. Their signs and node factors agree with Woodward's localized adiabatic identity. At either endpoint the distinguished equivariant insertion restricts through $\mathrm{Kir}_\pm$, hence becomes $[p_\pm]$ by (45); the quantum Kirwan derivatives remain inside $A_\pm$. Lemma 8.3 identifies the resulting endpoint covector with $c_{\dim Y_\pm}(z)^{-1}\mathfrak{r}_{Y_\pm,p_\pm}D\tau_{Y_\pm,-}$; differentiating the affine-gauged input gives the remaining factor $D\kappa_\pm$. The endpoints have the same dimension, so the common invertible scalar $c_{\dim Y_\pm}(z)$ cancels. Thus the coefficientwise Kalkman identity is (47) as an equality of functionals on the common equivariant input module. Applying arbitrary further input and bulk derivatives before coefficient extraction gives the polarized row identity, not only its value on the unit. □

For a birational map between smooth projective varieties, the required class exists globally. Włodarczyk's smooth birational cobordism admits a smooth projective equivariant completion whose source and sink quotients are the two endpoints [Wło00, Proposition 2(B')]. Choose p in the common birational open supplied as a trivialized cylinder by the construction. There the cobordism is a trivial orbit cylinder. The following lemma produces the class a_p from that orbit. It is stated for an arbitrary smooth projective $\mathbf{G}_m$-variety carrying the two chamber polarizations of Definition 8.1(i), so it applies directly to the resolved completion.

Lemma 8.5 (Orbit-cylinder disjointness). *Let W be a smooth projective variety with a $\mathbf{G}_m$-action and two equivariant ample linearizations L_- and L_+ at which stable equals semistable and the semistable loci are free. For a rational $u \in [0,1]$, write $L_u = L_-^{1-u} \otimes L_+^u$ for the interpolated polarization, and for a fixed point w write $\mu_u(w)$ for the weight of the $\mathbf{G}_m$-action on the fibre of L_u at w. Let $x \in W^{\mathrm{ss}}(L_-) \cap W^{\mathrm{ss}}(L_+)$, let $O = \mathbf{G}_m \cdot x$, and let $w_0 = \lim_{t\to 0} t \cdot x$ and $w_\infty = \lim_{t\to\infty} t \cdot x$ be its Białynicki–Birula limits in W. Then:*

(a) *$\overline{O} = O \cup \{w_0, w_\infty\}$, and $w_0 \neq w_\infty$ are fixed points;*

(b) *if a fixed component F is semistable for some L_u with $u \in [0,1]$, then $\overline{O} \cap F = \emptyset$;*

(c) *$\overline{O} \cap W^{\mathrm{ss}}(L_\pm) = O$, and the equivariant Poincaré dual $a = \mathrm{PD}[\overline{O}] \in H^*_{\mathbf{G}_m}(W)$ satisfies $\mathrm{Kir}_\pm(a) = [q_\pm]$, where $q_\pm$ are the images of x in the two quotients, and $a|_F = 0$ for every fixed component F as in* (b)*.*

Proof. We use the numerical criterion for $\mathbf{G}_m$ in the fibre-weight form: a point y is semistable for an equivariant ample M if and only if

$$\mu_M(y_\infty) \leq 0 \leq \mu_M(y_0), \tag{49}$$

where y_0, y_∞ are its two limits and μ_M is the fibre weight; y is stable if and only if both inequalities are strict. This is Mumford's numerical criterion [MFK94, Chapter 2, §2.1, Theorem 2.1] made explicit for the two one-parameter subgroups of $\mathbf{G}_m$: embed W equivariantly in $\mathbf{P}(V)$ by sections of a power of M. For $v = \sum_k v_k$ with $t \cdot v_k = t^k v_k$, the limit of $[t \cdot v]$ at $t \to 0$ is $[v_{k_{\min}}]$ and at $t \to \infty$ is $[v_{k_{\max}}]$, where $k_{\min} < k_{\max}$ bound the support of v; the fibre of $\mathcal{O}(1)$ at the fixed point $[v_k]$ has weight $-k$; and $[v]$ is semistable exactly when

$k_{\min} \leq 0 \leq k_{\max}$. A fixed point is its own pair of limits, so it is semistable for M exactly when μ_M vanishes on it.

(a) The stabilizer of x is trivial because the semistable loci are free, so $O \cong \mathbf{G}_m$ and its closure in the projective W is the image of the completed parametrization $\mathbf{P}^1 \to W$, $t \mapsto t \cdot x$; it adds exactly the two limits, which are fixed. Strict inequalities in (49) for L_- give $\mu_0(w_0) > 0 > \mu_0(w_\infty)$, so $w_0 \neq w_\infty$.

(b) The fibre weight $\mu_u(w)$ is affine in u. Stability of x at $u = 0$ and $u = 1$ gives $\mu_u(w_0) > 0$ at both ends, hence $\mu_u(w_0) > 0$ for every $u \in [0, 1]$; likewise $\mu_u(w_\infty) < 0$ on all of $[0, 1]$. A fixed component F semistable at L_u has $\mu_u(F) = 0$ by the criterion applied to its points. Since μ_u is constant on F, neither w_0 nor w_∞ lies in F. The open orbit O contains no fixed point, so $\overline{O} \cap F = \emptyset$.

(c) The limits are unstable for both $L_\pm$: their μ_u-values at $u \in \{0, 1\}$ are nonzero. Hence $\overline{O} \cap W^{\mathrm{ss}}(L_\pm) = O$. Restriction of a Poincaré dual to an open invariant subset is the Poincaré dual of the intersected cycle, so a restricts on $W^{\mathrm{ss}}(L_\pm)$ to $\mathrm{PD}[O]$. Under the free quotient this is the point class of the image of O, which is $[q_\pm]$. The vanishing $a|_F = 0$ follows from (b), since the restriction of $\mathrm{PD}[\overline{O}]$ to a closed subvariety disjoint from $\overline{O}$ is zero. □

Applied to the resolved completion, with x the cylinder point over p, the lemma produces the class a_p in (45). The cylinder orbit is free. It is semistable for both chamber polarizations because the identification of the extreme quotients with the two endpoints in Definition 8.1(i) is an identification of birational varieties: it restricts over the common open to the cobordism quotients, which are defined at p, and the free semistable fibre over $p_\pm$ is then the cylinder orbit itself. A fixed component occurring at an intermediate polarization in the wall-crossing sense is semistable for some interpolated L_u, so clause (b) covers every wall term in the polarization sweep. The lemma answers the completion question intrinsically: its hypotheses are conditions on the resolved completion itself, namely Definition 8.1(i) and bistability of the cylinder orbit, so no compatibility between the resolution and the orbit limits is needed. In particular it does not matter that the limits w_0, w_∞ lie outside the cylinder over which the resolution is an isomorphism; the numerical criterion is evaluated upstairs. This assertion concerns the ordinary Kirwan restrictions only; no vanishing of positive-degree corrections to the quantum Kirwan maps is used. The cited cobordism theorem supplies the underlying completion and cylinder; it does not by itself verify conditions (i)–(iii) of Definition 8.1. Those conditions remain explicit assumptions in the global theorem.

There is a terminology point in the source. Włodarczyk calls the open cobordism *projective* when it is quasiprojective. We use the resolved proper equivariant completion constructed in the proof of Proposition 2(B'): the equivariant projective completion of the glued double line-bundle space, followed by its canonical equivariant resolution. The glued space is smooth, so the canonical resolution is an isomorphism over it, in particular over the source, sink, and common-open cylinder. Thus the extreme quotients and the cylinder orbit are unchanged. Lemma 8.5 uses nothing else about the completion.

8.2 Neutral tails and threshold comparison

The two maps in (44) initially belong to opposite Novikov completions. The comparison needed for the point-row Boolean is smaller than a common field for the complete source. Fix a finite ample-energy and bulk Artin quotient of the equivariant Novikov ring. For a primitive affine gauge direction δ, all ordinary curve degrees and fixed graph types are then finite; only the integer coordinate along δ can remain unbounded. We first control those unbounded tails.

The argument separates the fixed geometry from the remaining analytic comparison. A rank-one derived-intersection theorem makes the fixed clutching data constant along each affine tail; the moving normal weights then give a Gamma recurrence. Marked threshold compatibility subsequently joins the resulting cyclic z-modules across the finitely many changes of sign, stability, and zero-mode rank.

Call the direction neutral when

$$c_1^{\mathbf{G}_m}(TW) \cdot \delta = 0. \tag{50}$$

We compare formal monodromy after the Rees substitution

$$Q^d \longmapsto z^{c_1^{\mathbf{G}_m}(TW)\cdot d} Q^d. \tag{51}$$

Proposition 8.6 (Gamma-ratio reduction)**.** *At any finite Artin level, every individual rotation-fixed graph coefficient in a neutral direction is a finite nilpotent derivative of a vector-valued balanced Gamma-ratio kernel. Its signed slope is $c_1^{\mathbf{G}_m}(TW)\cdot\delta$. In a nonneutral direction, every homogeneous coefficient is Laurent-finite.*

Proof. Woodward's classification of the rotation-fixed graph locus is exhaustive for sufficiently large area [Woo18, Corollary 9.10]. His virtual-normal splitting gives

$$\mathrm{Eul}(N_\pm) = \mathrm{Eul}((R\pi_* \,\mathrm{ev}^*\, T(W/\mathbf{G}_m))_{\mathrm{mov}})\,(\mp\zeta)(\mp\zeta - \psi). \tag{52}$$

The last two factors smooth the node and move the attaching point. They are independent of the affine gauge degree, so gluing neither removes one end of the degree lattice nor creates a degree-dependent pole. Formula (52) applies when a nontrivial attached component is present; in the irreducible unstable type those factors are absent and only the moving index remains [Woo18, Example 9.15].

Pass to a finite cover which clears stabilizer denominators and apply the splitting principle. A virtual line in the moving index of the principal weighted component has nilpotent Chern root α_a and degree

$$n_a(k) = h_a k + s_a. \tag{53}$$

For every integer n, its inverse index Euler factor is the single meromorphic expression

$$\frac{1}{\prod_{m=0}^{n}(\alpha_a + m\zeta)} = \zeta^{-n-1}\frac{\Gamma(\alpha_a/\zeta)}{\Gamma(\alpha_a/\zeta + n + 1)}. \tag{54}$$

For $n < 0$, the left side denotes the Euler class of H^1; for $n \geq 0$, it denotes the inverse Euler class of H^0. Thus the two degree rays are not assigned unrelated Gamma products.

The signed sum of the moving slopes is

$$\sum_a \epsilon_a h_a = c_1^{\mathbf{G}_m}(TW)\cdot\delta. \tag{55}$$

The gauge Lie algebra has slope zero. Moving modes on attached fixed bubbles have fixed rank at the chosen ordinary degree. Their Euler factors are finite products of linear functions of k, possibly inverted. Each inverse factor is an adjacent Gamma ratio

$$(hk + a)^{-1} = \frac{\Gamma(hk + a)}{\Gamma(hk + a + 1)}, \tag{56}$$

so it has zero signed slope and does not alter (55). Equivariant inputs are polynomial in k; Chern roots and psi classes are nilpotent at Artin level; Liouville classes supply the Fourier exponential.

Under (50), equations (52)–(56) therefore give a balanced Gamma-ratio kernel coefficientwise. Polynomial equivariant insertions are Fourier derivatives, while nilpotent Chern-root and bulk shifts give only finitely many parameter derivatives. For comparison, one moving-character residue has the exact normalization

$$\operatorname*{Res}_{h\sigma+\alpha=-m} \Gamma(h\sigma+\alpha)\, d\sigma = \frac{(-1)^m}{h\, m!}. \tag{57}$$

The factorial is the moving-mode Euler factor, $1/h$ is the one-character Jacobian factor, and the sign is the complex orientation.

If (50) fails, the virtual-dimension equation fixes k in each homogeneous coefficient, hence the coefficient is Laurent-finite. This proves the last assertion. □

Remark 8.7 (Higher-pole localization boundary). Equation (57) is used only to normalize one simple moving-character pole. Coincident or higher poles require derivatives and the full localization expression, including graph automorphism factors; no product-of-simple-residues formula is used or asserted.

Remark 8.8 (Exact analytic boundary). Aleshkin–Liu prove the relevant contour statement for a finite-dimensional linear abelian GLSM with adjacent secondary-fan chambers, a circuit, Calabi–Yau charges, and a grade-restriction window [AL23, Definition 5.18 and Theorem 5.21]. Proposition 8.6 does not construct those data for a nonlinear virtual fixed graph. Their theorem therefore supplies neither the derived-intersection argument below nor the marked threshold comparisons.

González–Woodward's crepant wall-crossing theorem gives a weaker distributional statement: neutral Picard shifts produce sums of derivatives of delta distributions [GW15, Remark 1.18(d), Remark 4.6, and equation (47)], and the authors explicitly do not deduce analytic continuation without an independent convergence input [GW15, Remark 4.7]. This does not replace Hypotheses 8.13 and 8.15. Proposition 6.1 exhibits the underlying obstruction: information supported at a Fourier boundary can reappear with full support after inverse transform, so almost-everywhere equality does not identify the marked formal packet.

Theorem 8.9 (Rank-one tailwise derived identification)**.** *For every finite Artin level, ordinary bubble degree, and rotation-fixed graph type, after splitting the affine degrees into stabilizer congruence classes and removing the finite sign, stability, and zero-mode thresholds, there are isomorphisms of the fixed clutching stacks and their universal domains which identify consecutive degrees in each remaining tail. These isomorphisms identify the evaluation maps and the fixed parts of the relative perfect obstruction theories as morphisms to the corresponding cotangent complexes, hence also their virtual classes. They respect stack automorphisms, gluing over the inertia quotient, and reduction between Artin levels. Stable degree-zero attached bubbles are retained as fixed stable-map factors; the irreducible unstable type is treated by its separate convention.*

Proof. Fix the generic semistable locus and the two endpoint fixed components specified by the graph type. Write W_F^a for the corresponding Białynicki–Birula stratum: its points lie in that generic locus and satisfy $\lim_{t\to 0} t^a x \in F$. Evaluation at 1 identifies fixed sections on one parametrized affine chart with W_F^a: the section determined by x is $z \mapsto z^a x$, and it extends across the origin with limit in F exactly when $x \in W_F^a$. A principal clutching section has two such extensions with the same generic value. Let $\mathfrak{S}_{a_-,a_+}$ denote the derived fixed section stack for the chosen clutching bundles, stabilizer lift, and graph component. Evaluation at 1 is a functorial equivalence on derived bases

$$\mathfrak{S}_{a_-,a_+} \simeq \mathfrak{Z} := [W_{F_-}^{a_-}/\mathbf{G}_m] \overset{\mathbf{R}}{\underset{[W/\mathbf{G}_m]}{\times}} [W_{F_+}^{a_+}/\mathbf{G}_m].$$

Here is an affine verification of the derived assertion. On a $\mathbf{G}_m$-invariant Sumihiro chart $\operatorname{Spec} A$, write $A = \bigoplus_w A_w$. For a derived test algebra R, an equivariant map from the parametrized affine chart is a graded morphism $A \to R[z]$; after evaluation at 1, a homogeneous element of weight w has the unique extension $f \mapsto f(x)z^{aw}$. Regularity at the origin says exactly that the coefficients with $aw < 0$ vanish. Thus the functor factors through the derived base change of the attractor algebra $A/(A_w : aw < 0)$, with the component and semistability open imposed. This calculation is valid for simplicial commutative test algebras degreewise, commutes with base change, and glues over the equivariant affine cover. After quotienting by change of trivialization it proves the one-chart equivalence with $[W_F^a/\mathbf{G}_m]$, including stabilizers and nilpotent derived families. Derived descent over the two-chart cover then gives the displayed homotopy fibre product. Its inverse equips a family x with the prescribed two clutching bundles and the sections $z \mapsto z^{a_\pm}x$. The chosen strata impose the generic stability and graph-component conditions. Appendix A.2 carries out this affine verification, together with chart independence and the descent step. The attached stable-map factors in Woodward's equations (54)–(57) are added by derived fibre products over the two inertia evaluations.

For $\mathbf{G}_m$, replacing a nonzero endpoint exponent by another integer of the same sign leaves the relevant stratum and limit map unchanged. Passing to a fixed stabilizer congruence class also fixes the inertia label. Thus the displayed derived intersection, its universal domain and evaluations, and its derived fibre products with the fixed ordinary stable-map factors are identical throughout one tail. The universal gauged maps $z \mapsto z^a x$ themselves are not identified when a changes; it remains to compare the fixed deformation complexes.

All tangent and cotangent complexes in the next calculation are relative to the fixed moduli of marked domains, denoted $\mathfrak{M}$. The tangent complex of the displayed derived intersection is

$$\left[T(W_{F_-}^{a_-}/\mathbf{G}_m) \oplus T(W_{F_+}^{a_+}/\mathbf{G}_m) \longrightarrow T(W/\mathbf{G}_m)\right].$$

It retains the degree-one term when the two attracting loci meet nontransversely. Equivalently, trivialize the pulled-back tangent complex on the open orbit by evaluation at 1. Taking invariants in the two-chart Čech resolution gives

$$\left[F_{\leq 0}^{0}V \oplus F_{\leq 0}^{\infty}V \longrightarrow V\right],$$

where the two terms are the extension subbundles at the endpoints. They are the tangent bundles of the two attracting loci. Thus this invariant Čech complex agrees with the preceding tangent complex without choosing an equivariant splitting. More precisely, in perfect complexes on $\mathfrak{Z}$, functoriality of restriction to the two charts gives

$$(R\pi_* u_a^* T(W/\mathbf{G}_m))^{\mathrm{rot}} \simeq \left[F_{\leq 0}^{0}V \oplus F_{\leq 0}^{\infty}V \longrightarrow V\right] \simeq T_{\mathfrak{Z}/\mathfrak{M}}.$$

The composite is the differential of evaluation at 1: on the two-chart cover, both it and the tangent map of the derived fibre product are induced by the same restriction morphisms to the open orbit. This identifies the complex together with its canonical map to the tangent complex, not only its class in K-theory. Along a tail the signs of all endpoint weights are constant, and the zero-weight changes were removed as thresholds. The two extension subbundles and the restriction morphism are therefore constant, even though the universal gauged map varies with a.

Put $E_a = ((R\pi_* u_a^* T(W/\mathbf{G}_m))^{\mathrm{rot}})^\vee$. Woodward's relative perfect obstruction theory is the natural truncation map $E_a \to L_{t_0\mathfrak{S}_{a_-,a_+}/\mathfrak{M}}$ [Woo18, Definition 7.13]. The mapping-stack cotangent formula identifies E_a with $\tau_{[-1,0]}L_{\mathfrak{S}_{a_-,a_+}/\mathfrak{M}}$. Under the derived equivalence

$\mathrm{ev}_1 : \mathfrak{S}_{a_-,a_+} \simeq \mathfrak{Z}$, the comparison is the following 2-commutative square:

$$\begin{array}{ccc} E_a & \longrightarrow & L_{t_0\mathfrak{S}_{a_-,a_+}/\mathfrak{M}} \\ \downarrow\sim & & \downarrow\sim \\ \tau_{[-1,0]}L_{\mathfrak{Z}/\mathfrak{M}} & \longrightarrow & L_{t_0\mathfrak{Z}/\mathfrak{M}}. \end{array}$$

The lower arrow is the canonical truncation map. The square is the naturality square of the truncation transformation evaluated at the equivalence ev_1, and on the two-chart Čech presentation both composites are the same difference of restrictions to the open orbit, so the comparison homotopy may be taken to be the identity (Proposition A.10). The left vertical arrow is therefore a quasi-isomorphism, and the two perfect obstruction theories are isomorphic as morphisms to the cotangent complex, not merely as perfect complexes. Appendix A.3 also records the two imported comparisons the argument uses.

For an attached stable-map factor, the fixed graph stack is a derived fibre product over its inertia evaluation. Its cotangent triangle is the direct sum of the two factor obstruction theories with the pullback of the inertia cotangent complex removed. The comparison above is the identity on the stable-map and inertia terms, so it gives an isomorphism of these exact triangles. This is the perfect-obstruction-theory compatibility used in Woodward's cutting formula [Woo18, Proposition 7.14], and it identifies the virtual classes.

On a k-fold orbifold chart the same calculation takes place on $[\operatorname{Spec} R[z^{1/k}]/\mu_k]$. In characteristic zero, μ_k-invariants are exact. Fixing the stabilizer congruence fixes the graded universal bundle and its twisted-sector evaluation, so the preceding equivalence and the 2-commutative obstruction-theory square are μ_k-equivariant. Rigidification quotients both sides by the same finite central gerbe; pullback to that gerbe is conservative, and hence the equivalence and obstruction theory descend. Stabilizer and graph-automorphism groups, including their localization multiplicities, are unchanged. The parametrized principal component has no domain automorphisms, while the fixed bubble factors retain theirs. Appendix A.4 expands the orbifold chart, the invariants, and the descent along the gerbe.

For the irreducible unstable convention there is no stable-map factor. Its fixed deformation complex is exactly $[F^0_{\le 0}V \oplus F^{\infty}_{\le 0}V \to V]$; the moving complement is the normal index, and the node and attaching factors are absent. Thus the same square applies without a gluing term. Every construction above is a finite derived base change, so it commutes with Artin reduction. This proves every assertion. ☐

Proposition 8.10 (Clutching-tail holonomicity)**.** *Every point-marked endpoint tail is a finite nilpotent derivative of one Gamma-ratio recurrence with affine arguments. Its generating series is holonomic over the finite Artin coefficient algebra. In a neutral direction its coefficients have polynomial-logarithmic growth after one exponential rescaling, so the tail has radial boundary values which are distributions of finite order.*

Proof. Corollary 9.10 and equations (54), (56), and (57) of [Woo18] express a rotation-fixed component as a fibre product of two clutching factors over the inertia quotient. Each factor already contains the complete ordinary stable-map space attached at that end. Thus arbitrary bubble trees occur in fixed proper stable-map factors once their ordinary degrees and markings are fixed; no induction over bubble-tree edges is required. Theorem 8.9 supplies the degree-shift identification of those fixed factors, their obstruction theories, and their evaluations. This is stronger than the normal-complex identity in equations (58)–(59) of [Woo18].

For $G = \mathbf{G}_m$, an affine clutching cocharacter is an integer. On one of the identified tails, apply the splitting principle to the moving part of the normal complex. A character root of weight h has degree $hk + s$, so the complementary moving weights give (54), with nilpotent Chern roots on the fixed coefficient stack. Stable attached components, including degree-zero bubbles stabilized by markings and nodes, retain the node and attaching factors in (52). Only the irreducible unstable type has no such factors. They are adjacent Gamma ratios and do not change the signed slope.

Only finitely many parameter derivatives are therefore needed. Gamma arguments do not cross a pole inside a tail, since every zero-mode crossing was removed as a threshold. Clearing recurrence denominators is therefore legitimate there. If $c_k \in R_N^r$ is the resulting virtual pushforward, Gamma recurrence gives a rational finite-state recurrence in k. Clearing its denominators yields

$$\sum_{i=0}^{d} p_i(k)c_{k+i} = 0, \qquad p_i(k) \in R_N[k],$$

on the tail. Hence $F(x) = \sum_{k\geq k_0} c_k x^k$ is (D)-finite over R_N. Finite Chern-root, input, and bulk derivatives commute with this identity and preserve (D)-finiteness. Under (50), Stirling's formula cancels the $k \log k$ terms in the signed product; the remaining exponential is removed by one rescaling and the coefficients grow by a power of k and $\log k$. After that rescaling, radial limits of F therefore define finite-order distributions, with polynomial-logarithmic bounds in the radial parameter. This is the meaning of tempered radial boundary values here. Finite threshold terms are Laurent polynomials and do not alter the conclusion. □

Remark 8.11 (Two-tail threshold obstruction). Tailwise holonomicity does not identify the two selected solutions. The bilateral sequence

$$c_k = \begin{cases} 2^k, & k \geq 0, \\ 3^{-k}, & k < 0 \end{cases}$$

has oriented germs

$$F_0(x) = \frac{1}{1-2x}, \qquad F_\infty(x) = \frac{3}{x-3}.$$

Both tails are first-order hypergeometric and tempered on their natural boundary circles. Away from the single threshold, the bilateral sequence is annihilated by $(E-2)(E-1/3)$; multiplying by a polynomial in the index absorbs the finite defect. Nevertheless the two displayed rational functions are not branches of one meromorphic function. Thus neither a polynomial recurrence nor tailwise holonomicity determines the finite connection data across a threshold.

Fix a finite local Artin coefficient algebra R_N, a ramified admissible sector S, and the duality convention of Section 2. Let $\mathcal{C}_N(S)$ be the exact category of finite locally free dual Rees z-differential modules on S with good Stokes filtration and ramified deck action; morphisms are horizontal and strict for the Stokes and Rees filtrations. We use nearby and vanishing cycles only for the specified meromorphic families in Hypothesis 8.15; their existence, strictness, and finite local freeness are part of that hypothesis. They are taken after passage to the dual category, so our convention is $\mathrm{var} : \phi_t M \to \psi_t M$; dualizing first exchanges the usual canonical and variation maps.

Definition 8.12 (Finite dual cyclic Rees module)**.** If $L \subset H$ is a submodule of an ambient Rees–Stokes module, its *strict Rees–Stokes saturation* is

$$\mathrm{Sat}_H^{\mathrm{str}}(L) := H \cap L[z^{-1}] \subset H[z^{-1}],$$

equipped with the induced Stokes filtration $F_\alpha \operatorname{Sat}_H^{\mathrm{str}}(L) = \operatorname{Sat}_H^{\mathrm{str}}(L) \cap F_\alpha H$. Whenever this operation is used below, finite local freeness of the result and strictness of its inclusion are part of Hypotheses 8.13 and 8.15.

Let T be normalized formal monodromy on an ambient object of rank d, let G be its finite ramified deck group, and let r be the marked horizontal row. A *finite dual cyclic Rees z-differential module* is the strict Rees–Stokes saturation, inside that ambient object, of the finite submodule generated by

$$\{g(rT^k) : g \in G,\ 0 \le k < d\}.$$

The threshold hypotheses require this saturation to be finite locally free. Cayley–Hamilton makes it stable under T, and it inherits the connection; it is not generated by applying the connection to a horizontal section. Its distinguished formal-monodromy Krylov subspace is $C_T(r)$ from Lemma 8.18.

Hypothesis 8.13 (Marked threshold compatibility: sign and stability thresholds)**.** At every finite Artin level and fixed ordinary degree, choose a common ramified admissible z-sector. At a sign or stability threshold, let K_j^- and K_j^+ be the finite dual cyclic Rees z-differential modules of the two adjacent tails. There must be a simply connected local domain U_j in the Fourier variable, with oriented cuts fixed, and one marked finite locally free Rees module $(\mathcal{K}_j, \mathfrak{r}_j)$ on $U_j \times S$. It carries compatible integrable connections in the Fourier and z directions, a relative good Stokes filtration, and deck action; its two oriented boundary restrictions are the adjacent cyclic modules with their marked rows. Choose an oriented path $\gamma_j \subset U_j$ between those restrictions. The threshold map is parallel transport along γ_j; it is not an arbitrary isomorphism of two abstract cyclic modules. The boundary identifications use the common equivariant-input module on which (47) is stated and are the identity on its marked coefficient functions and their chosen input and bulk derivatives; a wall-dependent change of input frame is not allowed. Remark 8.11 explains why tailwise holonomicity cannot replace this one-object condition.

Denote by $r_j^\pm$ the marked rows and by $T_j^\pm$ the half-Tate-normalized formal monodromies of $K_j^\pm$. The one-object continuation induces an invertible comparison

$$\Phi_j : K_j^- \xrightarrow{\sim} K_j^+, \qquad \Phi_j T_j^- = T_j^+ \Phi_j, \qquad \Phi_j(r_j^-) = r_j^+. \tag{58}$$

It is horizontal for the tail and z-connections, preserves the chosen Stokes filtration, is equivariant for the ramified deck action, and commutes with input and bulk derivatives, quotient maps between Artin levels, and the Rees substitution (51). It identifies the marked boundary germs of the two adjacent holonomic tails constructed in Proposition 8.10. These maps are compatible over the complete separated inverse system of Artin quotients.

Definition 8.14 (Reduced nearby cycles of a marked meromorphic family)**.** Let $\mathcal{M}$ be a specified marked meromorphic Rees family in a transverse parameter t, with ramified Stokes-filtered nearby- and vanishing-cycle functors in the standard triangle

$$i^*\mathcal{M} \longrightarrow \psi_t\mathcal{M} \xrightarrow{\mathrm{can}} \phi_t\mathcal{M} \xrightarrow{+1}, \qquad \mathrm{var} : \phi_t\mathcal{M} \longrightarrow \psi_t\mathcal{M}.$$

Let d be the rank of $\psi_t\mathcal{M}$ and let $V_t(\mathcal{M}) \subset \psi_t\mathcal{M}$ be the strict Rees–Stokes saturation of the finite submodule generated by

$$\{g(T^k v) : g \in G,\ 0 \le k < d,\ v \in \operatorname{im}(\mathrm{var})\}.$$

Cayley–Hamilton gives stability under formal monodromy, while the full G-orbit in the generators gives deck stability. The *reduced nearby cycles* of $\mathcal{M}$ are

$$\psi_t^{\mathrm{red}}\mathcal{M} := \psi_t\mathcal{M}/V_t(\mathcal{M}).$$

Thus reduced nearby cycles are defined by the nearby/vanishing-cycle triangle, not by a choice of a complementary summand. The usual canonical/variation relation makes $V_t(\mathcal{M})$ the canonical monodromy–deck-stable part generated by vanishing variation. Whenever this construction is used below, strictness of the inclusion $V_t(\mathcal{M}) \subset \psi_t\mathcal{M}$ and finite local freeness of the quotient over R_N are part of Hypothesis 8.15; functoriality of var then makes the induced monodromy and deck actions well defined.

Hypothesis 8.15 (Marked threshold compatibility: zero modes)**.** At a zero-mode rank change, choose a transverse parameter t and a specified marked meromorphic Rees family $\mathcal{M}_j$ whose two punctured sectorial boundary germs are the adjacent modules M_j^- and M_j^+, with the same fixed equivariant-input coordinate identifications. The inclusion $V_t(\mathcal{M}_j) \subset \psi_t\mathcal{M}_j$ of Definition 8.14 must be strict, and the reduced nearby cycles $\psi_t^{\mathrm{red}}\mathcal{M}_j$ must be finite locally free over R_N. Let $K_j^\pm$ be the strict Rees–Stokes saturations of the finite monodromy–deck orbits of the two specialized marked rows in the single object $\psi_t^{\mathrm{red}}\mathcal{M}_j$. Let $\widetilde{K}_j^\pm \subset M_j^\pm$ be the row-generated cyclic modules on the adjacent tails. Specialization is required to induce strict isomorphisms

$$\mathrm{sp}_j^\pm : \widetilde{K}_j^\pm \xrightarrow{\sim} K_j^\pm \tag{59}$$

in $\mathcal{C}_N(S)$. These isomorphisms carry the marked rows to their specializations and intertwine the z-connection, formal monodromy, Stokes filtration, and deck action. Thus no statement about an individual primary projection is included in the zero-mode assumption.

Denote by $r_j^\pm$ the specialized marked rows and by $T_j^\pm$ the half-Tate-normalized formal monodromies in $\psi_t^{\mathrm{red}}\mathcal{M}_j$. The one-object continuation through $\mathcal{M}_j$ must induce an invertible comparison $\Phi_j : K_j^- \xrightarrow{\sim} K_j^+$ satisfying (58) together with every property listed after it in Hypothesis 8.13. Here Φ_j compares the two row-generated reduced nearby-cycle modules, not ambient modules of unequal rank. Together with (59), it therefore gives a strict isomorphism between the two adjacent row-generated cyclic modules. These maps are compatible over the complete separated inverse system of Artin quotients.

We refer to Hypotheses 8.13 and 8.15 together as *marked threshold compatibility.*

At a zero mode, the unproved assertion therefore has two parts: construct the marked meromorphic threshold family, and prove that its vanishing-cycle submodule removes no part of the row-generated cyclic module. The second part is exactly the strict specialization isomorphism (59).

Condition (58) concerns only the cyclic module and allows arbitrary connection on the complementary source. The cleanest assertion known to us that would imply marked threshold compatibility is a one-object Gamma/window comparison. We record it as a conjecture; it is new to this manuscript, not a theorem from the literature.

Conjecture 8.16 (Marked Gamma/window continuation)**.** *At every finite Artin level, ordinary degree, neutral direction, and threshold of a gauged-admissible completion, the local Fourier object of Hypothesis 8.13, respectively the marked meromorphic family of Hypothesis 8.15, exists and admits a window presentation: in the fixed common equivariant-input frame, its two oriented boundary restrictions are generated by the Gamma-framed sections of one common window of topological K-classes; the window contains the skyscraper class of the marked common-open point, whose Gamma section induces the two marked rows under the flat pairing; the two boundary identifications of the window differ by a mutation whose cone is generated by classes supported on the wall; and, at a zero mode, the vanishing-cycle submodule $V_t(\mathcal{M}_j)$ is generated by the Gamma sections of wall-supported window classes.*

Sketch of implication. What follows indicates why Conjecture 8.16 would imply both hypotheses. It is motivation rather than a proof, and nothing later in the paper depends on it. A window mutation is supported on the wall, whereas the skyscraper of a point in the common open is unchanged by the mutation and Euler-orthogonal to wall-supported objects, so the induced one-object continuation carries the marked row. At a zero mode, wall-supported generation makes $V_t(\mathcal{M}_j)$ Euler-orthogonal to the row-generated module, and nondegeneracy of the flat pairing on the window span then separates the two; this is what the strict specialization isomorphisms (59) require.

Pairing preservation and formal exponential labels alone do not imply (58), as Sections 5–7 show.

The primary Boolean does not imply marked threshold compatibility. Equality of the zero or nonzero primary projections supplies neither a horizontal map of Rees modules nor compatibility with Stokes filtrations, deck actions, boundary germs, or the Artin inverse system. Thus the hypotheses are independently falsifiable marked irregular-connection statements.

Remark 8.17 (Verification status and obstruction). No instance of marked threshold compatibility is proved in this paper for a geometric threshold of a projective master; Remark 8.8 records the analytic boundary. Two linear-toric calibrations are known, and the obstruction to going further can be stated exactly.

There is a geometric calibration in the linear toric model. For a torus acting on a vector space, Woodward identifies the localized gauged graph potential with Givental's I-function [Woo18, Example 9.15]. Iritani constructs one global Landau–Ginzburg Brieskorn module whose chamber completions are the toric quantum D-modules and, for weak-Fano toric blowups, identifies its sectorial decomposition with the Gamma-framed ambient and residual K-groups [Iri20, Theorems 1.3, 7.25, 7.31 and 7.33]. For a point in the common open set, its skyscraper class lies in the ambient summand and its Euler row annihilates every residual summand. Hence the analytic one-object and marked-row clauses have a genuine linear toric-gauged model induced by one Fourier object; the blowup $\mathrm{Bl}_p\,\mathbf{P}^2 \to \mathbf{P}^2$ is the smallest example. In its rank-two affine presentation with weights $(1,0),(1,0),(1,1),(0,1)$, the common marked input is already visible in ordinary equivariant cohomology: if x, y are the two character classes, then the chamber maps are $(x,y) \mapsto (H,0)$ and $(x,y) \mapsto (H-E,E)$. Thus $x(x+y)$ maps to the point class in both chambers and restricts to zero on the intermediate fixed quotient [Woo17, Example 5.23]. This discrepancy-one direction is nonneutral, so the example calibrates the one-object and marking conventions but is not an instance of the load-bearing neutral hypothesis. The ordinary-flop theorem in Section 4 gives the corresponding path-local neutral model; it does not identify an arbitrary threshold inside one global master. This does not verify marked threshold compatibility for the projective cobordism used here: one must still compare Woodward's clutching tails on that cobordism with the boundary modules of the toric mirror object, in the fixed common input-and-derivative frame required above. No such reduction-in-stages comparison is used below. In particular, merely relaxing projectivity of the master does not remove this gap: the affine presentation has a rank-two gauge torus, and passing to the rank-one cobordism is itself the missing common-frame comparison.

There is, however, a genuinely neutral linear-toric model. Across a crepant toric wall, Coates–Iritani–Jiang construct global quantum connections for the two chambers and a continuation gauge between them which intertwines the quantum and grading connections, preserves the pairing, and carries Gamma-framed flat sections by the Fourier–Mukai transform [CIJ18, Theorems 6.1 and 6.3]. Gluing along this gauge gives a local one-object calibration. After taking the non-equivariant specialization, the graph Fourier–Mukai transform sends

$\mathcal{O}_p$ to $\mathcal{O}_{p'}$ for a point in the common open. If Θ denotes the continuation gauge, its pairing compatibility and $\Theta s(\mathcal{O}_p) = s(\mathcal{O}_{p'})$ give

$$\mathfrak{r}_{X_+,p'}(\Theta v) = [\Theta v, \Theta s(\mathcal{O}_p)) = [v, s(\mathcal{O}_p)) = \mathfrak{r}_{X_-,p}(v).$$

Thus it carries the marked point row exactly. Together with Woodward's toric I-function formula, this verifies the intrinsic marked-continuation part of the ordinary, non-zero-mode mechanism at the QDM/I-function level in the neutral linear-toric model. It does not provide the inverse-system comparison for an arbitrary projective master or the reduced nearby-cycle assertion at a zero mode.

Lemma 8.18 (Cyclic-row spectral support). *Let T be an endomorphism of a finite dimensional vector space V, and let $r \in V^*$. Put*

$$C_T(r) = \operatorname{span}\{r, rT, rT^2, \ldots\} \subset V^*.$$

For every eigenvalue λ, the restriction of r to the generalized λ-eigenspace of T is nonzero if and only if the λ-primary part of $C_T(r)$ is nonzero.

Proof. Let $e_\lambda(T)$ be the polynomial Bézout projector onto the generalized λ-eigenspace. The relevant primary projection of the row is $re_\lambda(T)$. It belongs to $C_T(r)$ and vanishes exactly when the restriction of r does. □

8.3 The finite cyclic packet

Formal smooth-endpoint quantum Kirwan surjectivity is elementary in this setting. Modulo the positive-energy ideal, $D\kappa_\pm$ is the derivative of the classical Kirwan map and is surjective. Choose a linear section of the reduction. Its arbitrary lift has composite $1 + N$, with N in the augmentation ideal, and the convergent Neumann series $\sum_{j\geq 0}(-N)^j$ gives a right inverse. Since $D\tau_{Y_\pm,-}$ is an invertible fundamental solution, the maps (44) are surjective as well.

The Rees factor in (51) is needed to compare formal monodromy. Homogeneity of $D\kappa$ gives

$$\mu_Y A - A\mu_W = \frac{1}{2}A - \Theta_Q A, \tag{60}$$

because the acting group has dimension one and $\Theta_Q(Q^d) = c_1^{\mathbf{G}_m}(TW)\cdot d\, Q^d$. After (51), the Θ_Q-term becomes $z\partial_z$, so both endpoint maps intertwine the z-connections with the same half-Tate shift. The Euler-product term intertwines because the virtual-dimension identity makes the quantum Kirwan map homogeneous and its derivative a quantum-product morphism. The target bulk point $\kappa(0)$ has positive energy and is formally parallel to the large-radius point; this transport preserves the point row.

Choose finitely many common equivariant inputs whose images under each surjection in (44) span the corresponding endpoint module; take the union of two lift sets if necessary. Evaluation on these images embeds the dual endpoint module into one finite coordinate space. After (51), this embedding intertwines the dual z-connections up to the same half-Tate shift. Let $T_\pm$ denote the induced half-Tate-normalized formal monodromies. The Krylov spans

$$\operatorname{span}\{\mathfrak{r}_\pm, \mathfrak{r}_\pm T_\pm, \mathfrak{r}_\pm T_\pm^2, \ldots\}$$

are the distinguished formal-monodromy subspaces inside the finite dual cyclic modules in that coordinate space. Injectivity here is simply dual to the surjectivity of (44).

Proposition 8.4 identifies the marked coefficient functions before either endpoint completion is taken. Theorem 8.9 and Proposition 8.10 supply their holonomic tail systems, while nonneutral directions are Laurent-finite by Proposition 8.6. The following elementary gluing step records exactly what the remaining hypotheses add.

Lemma 8.19 (Finite marked threshold gluing)**.** *Fix R_N, an ordinary degree, and a neutral direction. Suppose the ordered affine line is cut into finitely many tails by its thresholds, and suppose the adjacent tail modules carry the maps in Hypotheses 8.13 and 8.15. Then their marked boundary germs glue to one object of $\mathcal{C}_N(S)$. Its endpoint realizations have the same zero or nonzero primary projections. These glued objects form an inverse system under $R_{N+1} \to R_N$.*

Proof. At a sign or stability threshold, glue by the strict horizontal isomorphism Φ_j. It identifies the marked germs, and polynomial functional calculus in (58) identifies every primary projection. At a zero-mode threshold, first apply the specified strict specialization isomorphism to reduced nearby cycles, use Φ_j there, and return through the inverse specialization isomorphism to the adjacent tail. This composite intertwines formal monodromy, so polynomial functional calculus gives the same zero or nonzero primary projections. There are finitely many thresholds at fixed R_N, so these maps have a finite ordered composite. Their compatibility with Artin reduction makes the composites an inverse system. No colimit of the complementary equivariant source is formed. □

Theorem 8.20 (Conditional birational invariance of the point-row primary Boolean)**.** *Let Y_- and Y_+ be smooth projective birational complex varieties. Suppose a Włodarczyk completion for this birational map is gauged-admissible in the sense of Definition 8.1 and satisfies Hypotheses 8.13 and 8.15 at every Artin level, ordinary degree, neutral affine gauge direction, and threshold. Then, for every formal-monodromy eigenvalue λ,*

$$\mathfrak{r}_{Y_-}|_{\mathcal{P}_\lambda(Y_-)} \neq 0 \quad \Longleftrightarrow \quad \mathfrak{r}_{Y_+}|_{\mathcal{P}_\lambda(Y_+)} \neq 0. \tag{61}$$

Here the packets are realized through the marked cyclic continuation in the hypothesis; no continuation of the complementary source is part of the statement.

Proof. Use the gauged-admissible completion and its orbit-cylinder class. Lemma 8.19 compares the two endpoint marked cyclic modules at every finite Artin level. More explicitly, polynomial functional calculus in (58) gives

$$\Phi_j\big(r_j^- e_\lambda(T_j^-)\big) = r_j^+ e_\lambda(T_j^+)$$

at every threshold. Hence the relevant primary row vanishes on one side if and only if it vanishes on the other. At a zero-mode threshold this equality is taken first in the row-generated reduced nearby-cycle modules and then transported to the adjacent tails by the strict isomorphisms (59). Apply Lemma 8.18 at the endpoints. Because the Artin inverse system is separated, a projected row is nonzero exactly when it survives at some finite level. This proves both implications. □

Under marked threshold compatibility, Theorem 8.20 is the global replacement for Hypothesis 7.1. Without those inputs, Remark 8.11 and the earlier countermodels show why the conclusion cannot be obtained from tailwise holonomicity, localized one-wall receivers, or unmarked Fourier boundary data alone.

9 The cubic endpoint

The conditional transport theorem contains no special feature of cubic threefolds. This section proves the endpoint statement needed for the application. We retain Cai's quantum-connection convention and reconstruct the exact block used below, so the application does not depend on translating a qualitative summary of his result.

9.1 The primitive-sixth block

Let $X \subset \mathbf{P}^4$ be a smooth cubic threefold, let P be its hyperplane class, and put $q = u^2 \neq 0$. In the basis $1, P, P^2, P^3$, Cai writes the small z-equation as

$$z^2 \partial_z S = (K + zG)S, \tag{62}$$

where

$$K = 2\begin{pmatrix} 0 & 6q & 0 & 36q^2 \\ 1 & 0 & 15q & 0 \\ 0 & 1 & 0 & 6q \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad G = \operatorname{diag}\left(\frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}\right). \tag{63}$$

See [Cai26, Section 3]. Over $\mathbf{Q}(\sqrt{3})(u)$, the eigenvalues of K are $6\sqrt{3}u, -6\sqrt{3}u, 0, 0$, and its zero eigenspace is one rank-two Jordan block. Solving the first off-block Sylvester equation gives, on this block,

$$M_1 = \begin{pmatrix} -19/18 & 0 \\ 0 & 19/18 \end{pmatrix}, \qquad (R_2)_{21} = -16/81, \tag{64}$$

when the Jordan link is normalized to one. The indicial polynomial is therefore

$$(\rho + 19/18)(\rho - 1/18) + 16/81 = \rho^2 + \rho + \frac{5}{36}. \tag{65}$$

Its roots are $-1/6$ and $-5/6$, hence the formal-monodromy eigenvalues of this block are ζ_6^{-1} and ζ_6. This direct calculation recovers the small-connection part of [Cai26, Proposition 6]. Cai's flatness argument in the same proposition gauges the big connection back to the small one and preserves this rank-two block.

9.2 The point coefficient

The preceding calculation finds the packet but does not say whether the point row detects it. That coefficient follows from one scalar period. Write

$$F(t) = \sum_{d \geq 0} \frac{(3d)!}{(d!)^5} t^d = {}_2F_3\left(\begin{matrix} 1/3,\ 2/3 \\ 1,\ 1,\ 1 \end{matrix}; 27t\right), \tag{66}$$

the equality of the two expressions being the identity $(3d)! = 27^d\, d!\, (1/3)_d (2/3)_d$.

Lemma 9.1 (Central charge of the point class)**.** *Let $X \subset \mathbf{P}^4$ be a smooth cubic threefold and $p \in X$. In the small quantum limit there are a nonzero constant C and an integer k, both independent of z and fixed by the chosen Gamma and graph-space normalizations, such that the component of the Gamma point section along the unit is*

$$Z_p(z) := (s_X(\mathcal{O}_p), \mathbf{1})_X = C\, z^{k-3/2} F(q/z^2). \tag{67}$$

Proof. Read (5) from the right. For a point on a smooth threefold $\operatorname{ch}(\mathcal{O}_p) = [p]$ with no lower-degree terms, and $[p]$ has cohomological degree 6, so $(2\pi i)^{\deg/2} \operatorname{ch}(\mathcal{O}_p) = (2\pi i)^3 [p]$. Every positive-degree term of $\widehat{\Gamma}_X$ annihilates the top class, so $\widehat{\Gamma}_X[p] = [p]$, and $c_1(X) \cup [p] = 0$ gives $z^{c_1(X)}[p] = [p]$. The grading operator satisfies $\mu[p] = (3 - \frac{3}{2})[p]$, whence $z^{-\mu}[p] = z^{-3/2}[p]$. Collecting these scalars,

$$s_X(\mathcal{O}_p) = (2\pi)^{-3/2} (2\pi i)^3 z^{-3/2}\, L_X(0, z)[p].$$

Pairing against the unit and expanding the fundamental solution in descendants leaves the one-point generating function with a point insertion, $\sum_d q^d\langle[p]/(z-\psi)\rangle_{0,1,d}$, which is the coefficient of $\mathbf{1}$ in the small J-function.

That coefficient is (66). Givental's mirror theorem for toric complete intersections gives the small I-function of a degree-three hypersurface in $\mathbf{P}^4$,

$$I(q,z) = z\sum_{d\geq 0} q^d \frac{\prod_{m=1}^{3d}(3P+mz)}{\prod_{m=1}^{d}(P+mz)^5}, \tag{68}$$

and identifies it with the small J-function up to a mirror map [Giv98, CG07]; the exponential prefactor $e^{(P\log q)/z}$ of the usual display is absorbed into the Novikov convention here, and is killed by the specialization $P=0$ below in any case. The degree-d term carries $z^{3d-5d}=z^{-2d}$, so $I=z\mathbf{1}+O(z^{-1})$ and the mirror map is trivial. This count uses the Fano index: index one would leave a z^0 term and a nontrivial mirror map, and a cubic threefold has index $5-3=2$, the smallest value for which the count closes. Setting $P=0$ in (68) leaves

$$z\sum_{d\geq 0} q^d\frac{(3d)!\,z^{3d}}{(d!)^5 z^{5d}} = z\,F(q/z^2),$$

which is that coefficient up to the power of z carried by the chosen normalization. The accumulated factors—$(2\pi)^{-3/2}(2\pi i)^3$, the sign convention relating $L_X(0,z)$ to J, and the overall power of z in the I- and J-function conventions—are a nonzero constant times an integral power of z. The one factor which could have contributed a further half-integral power does not: the normalization in (5) is $(2\pi)^{-\dim X/2}$, not $(2\pi z)^{-\dim X/2}$, so $z^{-\mu}$ is the only source of a fractional exponent. □

Only the fractional part of the exponent of z is used below, so the integer k in (67) never enters, and neither does a convention which replaces z by $-z$: that moves the branch phases but not the exponents.

The two asymptotic regimes of (67) match the two parts of (63). The eigenvalues $\pm 6\sqrt{3}u$ of K produce the exponentially large and small terms of the expansion, while the rank-two zero block—the one carrying the primitive-sixth formal monodromy—is regular singular and produces the algebraic terms. The packet in question is therefore read from the algebraic part alone.

That part is a Barnes expansion. For ${}_pF_q$ with $p=q-1$ and a_k distinct modulo the integers, its algebraic terms are

$$\frac{\prod_j\Gamma(b_j)}{\prod_j\Gamma(a_j)}\sum_k\frac{\Gamma(a_k)\prod_{j\neq k}\Gamma(a_j-a_k)}{\prod_j\Gamma(b_j-a_k)}\,(-x)^{-a_k}\,(1+O(x^{-1})) \tag{69}$$

[NIS26, Equations 16.11.2 and 16.11.8]. Here $a=(1/3,2/3)$ and $b=(1,1,1)$; the difference $1/3$ is not an integer, so no logarithms occur and the two algebraic series are distinct. The prefactor is $1/(\Gamma(1/3)\Gamma(2/3))$, and (69) gives the leading coefficients

$$\begin{aligned} C_{1/3} &= \frac{\Gamma(1/3)\Gamma(1/3)}{\Gamma(2/3)^3}\cdot\frac{1}{\Gamma(1/3)\Gamma(2/3)} = \frac{\Gamma(1/3)}{\Gamma(2/3)^4},\\ C_{2/3} &= \frac{\Gamma(2/3)\Gamma(-1/3)}{\Gamma(1/3)^3}\cdot\frac{1}{\Gamma(1/3)\Gamma(2/3)} = \frac{\Gamma(-1/3)}{\Gamma(1/3)^4}. \end{aligned} \tag{70}$$

The Gamma function has no zeros, and $-1/3$ is not one of its poles, so both coefficients are nonzero; the branch factors $(-x)^{-1/3}$ and $(-x)^{-2/3}$ are nonzero on every sector. Substituting $x=27q/z^2$ in (67) turns them into the powers

$$z^{-3/2}x^{-1/3}\sim z^{-5/6},\qquad z^{-3/2}x^{-2/3}\sim z^{-1/6}, \tag{71}$$

whose exponents reduce modulo the integers to the roots $-5/6$ and $-1/6$ of (65), with monodromy factors ζ_6 and ζ_6^{-1}. Thus the Gamma point section has a nonzero coefficient along each of the two formal lines of that block.

It remains to pass from the section to the row. The row is $v \mapsto [v, s_X(\mathcal{O}_p)]$, and (6) is nondegenerate; because of the half-turn $z \mapsto e^{-\pi i} z$ in its first slot, it pairs a formal line of monodromy λ with the line of monodromy λ^{-1}. The two lines of the block are inverse to each other, so the restriction of the pairing to that block is again nondegenerate. A nonzero coefficient of $s_X(\mathcal{O}_p)$ along one of the two lines therefore makes the row nonzero on the other. Both coefficients are nonzero here, so the conclusion is the same for both packets and does not depend on resolving which line is paired with which.

Proposition 9.2 (Cubic endpoint lemma)**.** *For every smooth cubic threefold X,*

$$\mathfrak{r}_X|_{\mathcal{P}_{\zeta_6}(X)} \neq 0, \qquad \mathfrak{r}_X|_{\mathcal{P}_{\zeta_6^{-1}}(X)} \neq 0. \tag{72}$$

The packet assertion follows from Cai's connection, while the point-row nonvanishing is the direct Barnes calculation (66)*–*(71)*. Both assertions are made at the formal-primary level: the sector-dependent Barnes phases prove nonvanishing of the intrinsic formal primary projections, and no comparison between alternative sectorial lifts is used in the endpoint contrast. The formal-primary Boolean is the one recovered from the cyclic row by Lemma 8.18. The paper uses half-Tate-normalized labels throughout. Relative to unnormalized monodromy, the common shift relabels both endpoints equally, so the normalized primitive-sixth contrast in* (72) *is unchanged.*

Remark 9.3 (Exact regression). The paper's verification package uses and verifies an explicit basis of $\operatorname{im} K^2 \oplus \ker K^2$, normalizes the nilpotent link, solves the first off-block Sylvester equation, and reconstructs (64)–(65) at three nonzero rational specializations of the quantum parameter by exact rational arithmetic. It also checks the hypergeometric recurrence behind (66) and the projective-space grading calculation below. The exact source, checker, and certificate are included in the archived release at `https://doi.org/10.5281/zenodo.21937490`; the artifact regresses the matrix-to-block and indicial arithmetic at three rational specializations, the hypergeometric recurrence, the Barnes non-pole arguments, and the projective-space grading identity. It does not verify the Barnes asymptotic theorem or quantum Künneth. The proof remains the calculation displayed above.

9.3 The two endpoints

For $\mathbf{P}^m$, write $n = m + 1$. At a nonzero quantum parameter $q_{\mathbf{P}}$, multiplication by $c_1(\mathbf{P}^m)$ is a weighted cyclic shift multiplied by n. After adjoining the n-th root of the parameter and rescaling the basis, it is $nq_{\mathbf{P}}^{1/n}$ times the standard cyclic shift. In its Fourier eigenbasis the diagonal of the grading operator is

$$\frac{1}{n}\sum_{k=0}^{m}(m/2 - k) = 0. \tag{73}$$

Indeed, if F is the unitary discrete Fourier matrix, then every diagonal entry of the Fourier conjugate of the grading operator is $\sum_j |F_{jk}|^2(m/2 - j) = n^{-1}\sum_j (m/2 - j)$, since $|F_{jk}|^2 = 1/n$. Thus the vanishing in (73) is entrywise, not merely a trace identity. All n rank-one formal factors therefore have formal-monodromy exponent zero. In particular, $\mathbf{P}^{m+3}$ has no primitive-sixth packet.

The quantum connection of a product is the tensor connection. Hence $X \times \mathbf{P}^m$ has $m+1$ copies of each cubic primitive-sixth line. The extended Gamma framing is multiplicative under products [Iri23, Remark 1.2], and the rank row is the tensor product of the two rank rows. Since the rank row of $\mathbf{P}^m$ is nonzero, (72) implies

$$\mathfrak{r}_{X\times\mathbf{P}^m}|_{\mathcal{P}_{\zeta_6}(X\times\mathbf{P}^m)} \neq 0, \qquad \mathcal{P}_{\zeta_6}(\mathbf{P}^{m+3}) = 0. \tag{74}$$

Proof of Theorem 1.1. If $X \times \mathbf{P}^m$ were rational, it would be birational to $\mathbf{P}^{m+3}$. Theorem 8.20 would identify the two point-row Booleans at ζ_6. Equation (74) gives the values one and zero, a contradiction. ☐

The separation between Theorems 8.20 and 1.1 is deliberate. The conditional transport argument does not use Cai's calculation. Any replacement endpoint with a point-row-detected primary packet gives another application of the same criterion.

10 Scope and source boundaries

The proof combines statements of different kinds. We record their logical roles explicitly.

(1) Theorem 3.2 is an exact identity inside the Gu–Yu–Yu completed formal comparison for $r_+ < r_-$. It concerns only the ambient point coordinate. It is not used to compose the global cobordism.

(2) Theorem 4.1 is an intrinsic point-section identity on one fixed Lee–Lin–Qu–Wang continuation domain. It does not assert full descendant invariance or a Gamma/Fourier–Mukai square for all classes.

(3) Corollary 3.5 requires its named one-wall sectorial realization hypothesis. The formal Gu–Yu–Yu decomposition and the extremal Shen–Shoemaker asymptotics do not by themselves identify the intrinsic large-radius point section in a common receiver.

(4) Propositions 5.1 and 6.1 are exact analytic and algebraic countermodels. They are not asserted to arise from smooth projective quantum connections. They show why the local receivers cannot simply be glued by formal monodromy, pairing, or localized support.

(5) Proposition 8.4 uses the global orbit-cylinder marking and every condition in Definition 8.1. Włodarczyk supplies the smooth completion and orbit cylinder, but the proper-DM master-stack, obstruction- theory, stability, and numerical-separation conditions remain assumptions. Lemma 8.5 proves the disjointness and Kirwan restrictions of the marked class from those stability assumptions and bistability of the cylinder orbit alone. The wall contributions vanish because their attaching nodes land in intermediate fixed loci on which this particular class restricts to zero. No claim is made that arbitrary insertions enjoy the same collapse.

(6) Proposition 8.6 proves the coefficientwise Gamma-ratio and balance identities for individual fixed-graph contributions. Theorem 8.9 supplies the degree-shift identification of the fixed stacks, obstruction theories, virtual classes, and evaluations. Consequently, Proposition 8.10 proves holonomicity and tempered growth for each complete clutching tail, including arbitrary bubble trees.

(7) Hypotheses 8.13 and 8.15 form an inverse-system family over all finite Artin levels, ordinary degrees, neutral directions, and thresholds. They ask for one-object local Fourier comparisons inducing strict isomorphisms of finite cyclic Rees z-modules

which intertwine formal monodromy, carry the point row, and at zero modes identify the entire adjacent row-generated module with its reduced nearby-cycle realization. Preservation of primary support is a consequence, not part of the hypotheses. No instance is proved here for a general geometric threshold; Remark 8.17 records the two linear-toric calibrations and the exact obstruction, and Conjecture 8.16 is the cleanest implying statement we know. It is new to this manuscript and unproved. Aleshkin–Liu prove a balanced contour theorem for finite-dimensional linear abelian GLSMs; they prove neither this marked compatibility nor its nonlinear virtual fixed-graph analogue. Remark 8.11 shows that separate holonomicity does not determine the threshold maps.

(8) The global proof continues only the finite cyclic module generated by the point row under formal monodromy. It does not assume that the entire equivariant quantum source has finite rank or admits a common continuation. No formal Novikov variable is evaluated at a nonzero complex number.

(9) Cai enters only Proposition 9.2. Equations (62)–(65) reconstruct the formal block from his displayed matrices, and (66)–(71) independently compute the point coefficient. That computation imports one further source: Givental's mirror theorem supplies the small I-function (68), and its mirror map is trivial here by the degree count in Lemma 9.1. The birational conditional transport theorem is unchanged if this endpoint lemma is replaced by any other detected primary packet.

Birational cobordism can have intermediate coarse quotients with finite quotient singularities [Wło00, AKMW02]. The global proof does not form their quantum D-modules. It uses only the smooth projective equivariant completion, the smooth extreme quotients, and the complete fixed loci in the gauged theory. This is why resolving and composing the intermediate walls is unnecessary.

Theorem 8.20 is conditional on both the gauged-admissibility conditions of Definition 8.1 and the inverse-system family in Hypotheses 8.13 and 8.15. It concerns the point row on a formal primary packet in the marked cyclic continuation postulated by those hypotheses. It is not a claim that all Stokes matrices, Gamma lattices, or derived categories are birational invariants. The incomplete-Gamma and punctual countermodels rule out those stronger inferences. What survives is exactly one Boolean row, obtained from one globally marked class.

The computational artifact has a similarly narrow role. It recomputes the cubic block and indicial polynomial, the period recurrence, and the projective grading identity. It does not verify virtual localization, the rank-one derived clutching theorem, or marked threshold compatibility. In particular, it cannot turn the conditional birational theorem into an unconditional one.

A The one-chart calculation

This appendix expands the proof of Theorem 8.9 at the three places where it is compressed: the affine verification of the derived equivalence, the 2-commutative obstruction-theory square, and the μ_k-equivariance and rigidification step. The hypotheses, the statement, and the conclusions of that theorem are unchanged; what follows adds no new assumption and asserts nothing stronger.

One convention must be fixed before any comparison. Woodward works with classical stacks carrying perfect obstruction theories in the sense of Behrend–Fantechi, over Artin stacks of prestable domains, and with Vistoli's bivariant Chow groups for representable morphisms; his index complex is $(Rp_*e^*T(X/G))^\vee$, stated absolutely in [Woo18, Definition 7.13(a)] and relative over the twisted domain stack in [Woo18, Section 8.3] and [Woo17, Example 6.6(c)]. Derived stacks play no role there. The derived fixed-section stacks used in Theorem 8.9 are an auxiliary layer of the present manuscript. Their only purpose is to make the degree-shift comparison functorial in nilpotent and simplicial thickenings of the test base, which is what upgrades an isomorphism of perfect complexes to an isomorphism of obstruction theories *as morphisms to a cotangent complex.* Every comparison below is returned to Woodward's setting by Proposition A.10: the classical truncation of the derived obstruction theory is his relative perfect obstruction theory over the fixed moduli of marked domains, so the identification is an identification of the relative virtual classes he uses.

The imported statements are: the clutching description of the rotation-fixed gauged maps [Woo18, Definition 9.7, Lemmas 9.8 and 9.9, Corollary 9.10]; the obstruction theory [Woo18, Definition 7.13] in its relative form [Woo18, Section 8.3], [Woo17, Example 6.6(c)]; the cutting axiom [Woo18, Proposition 7.14(b)] together with the fibre product over the diagonal on which its proof rests [Woo17, Proposition 5.21]; and the rigidified inertia conventions [Woo17, Section 4.3, Proposition 4.3(f),(g)]. Two standard comparisons are used and not reproved: the cotangent complex of a derived mapping stack is the pushforward of the pulled-back cotangent complex of the target, and a classical obstruction theory built from the deformation theory of a universal object is the classical truncation of the derived one. Remark A.13 says exactly where each enters.

A.1 Graded setup and the derived fixed-section stack

Throughout, W is the smooth projective $\mathbf{G}_m$-variety of Definition 8.1 and we work over $\mathbf{C}$. Fix the generic semistable locus $W^{\mathrm{gen}} \subset W$ and the two fixed components F_-, F_+ prescribed by the rotation-fixed graph type, as in the proof of Theorem 8.9. The gauge group is $\mathbf{G}_m$; the domain rotation group is written $\mathbf{C}^\times_{\mathrm{rot}}$.

The domain of the principal component is the parametrized line P with its two marked points $0, \infty$ and the two affine charts

$$P^0 = \operatorname{Spec} \mathbf{C}[z], \qquad P^\infty = \operatorname{Spec} \mathbf{C}[z'], \qquad z' = z^{-1},$$

glued along the open orbit $\mathbf{G}_m \subset P$. Each chart carries the rotation action scaling its own coordinate. We record an exponent $a_\pm \in \mathbf{Z}$ in each chart's own coordinate, so that both extension conditions below are conditions at the vanishing locus of the local coordinate; this matches the convention $W^a_F = \{x \in W^{\mathrm{gen}} : \lim_{t\to 0} t^a x \in F\}$ used in the proof of Theorem 8.9.

Definition A.1 (Derived fixed-section stack)**.** Let R be a simplicial commutative $\mathbf{C}$-algebra. An R-point of $\mathfrak{S}_{a_-,a_+}$ is a rotation-equivariant map $u\colon P_R \to [W/\mathbf{G}_m]$ whose rotation-equivariant structure on the underlying $\mathbf{G}_m$-bundle is the clutching datum of [Woo18, Definition 9.7] for the pair of cocharacters $t \mapsto t^{a_\pm}$, together with the prescribed stabilizer lift and graph component. Morphisms of R-points are isomorphisms of these data. This is a functor on simplicial commutative $\mathbf{C}$-algebras, defined degreewise in the simplicial direction; characteristic zero is used throughout, both for simplicial commutative algebras to model connective commutative differential graded algebras and for exactness of invariants under the reductive groups that appear.

By Sumihiro's theorem W is covered by $\mathbf{G}_m$-invariant affine opens. Fix one, $\operatorname{Spec} A$, and write its coordinate ring as a $\mathbf{G}_m$-representation,

$$A = \bigoplus_{w\in\mathbf{Z}} A_w, \qquad f \in A_w \iff f(t\cdot x) = t^w f(x). \tag{75}$$

Over a test algebra R, the trivialized chart at 0 has coordinate ring $R[z]$, graded with z in degree 1 and R in degree 0; the open orbit has $R[z^{\pm 1}]$ with the same grading.

Remark A.2 (What the exponent does and does not change). The exponent a enters the chart only through the twist of the rotation-equivariant structure on the $\mathbf{G}_m$-bundle. It does not change the domain, the target, or the trivializations. This is the reason the calculations below separate cleanly into a part that depends on a only through its sign and a part—the universal section itself—that depends on a exactly.

A.2 Evaluation at one and the attractor algebra

Lemma A.3 (Unique extension on one chart)**.** *Let R be a simplicial commutative $\mathbf{C}$-algebra and $a \in \mathbf{Z}$.*

(a) *Ring maps $\psi\colon A \to R[z^{\pm 1}]$ sending A_w into the z-degree aw part — the equivariance condition for a rotation-fixed section with clutching exponent a, for the gradings* (75) *— correspond bijectively, degreewise in the simplicial direction, to ring maps $x\colon A \to R$, by*

$$\psi(f) = x(f)\, z^{aw} \qquad (f \in A_w). \tag{76}$$

The correspondence is evaluation at $z = 1$.

(b) *Such a ψ takes values in $R[z]$ if and only if x annihilates every A_w with $aw < 0$, that is, if and only if x factors through the* attractor algebra

$$A^a := A/I_a, \qquad I_a := (A_w : aw < 0). \tag{77}$$

Proof. The grading forces $\psi(A_w) \subset (R[z^{\pm 1}])_{aw} = R\, z^{aw}$, so ψ is determined by the R-linear maps $x_w\colon A_w \to R$ defined by $\psi(f) = x_w(f) z^{aw}$. Since $z^{aw} z^{aw'} = z^{a(w+w')}$, the collection (x_w) is multiplicative if and only if the single map $x = \bigoplus_w x_w\colon A \to R$ is a ring map; unitality corresponds to $x(1) = 1$. This is (a), and both directions are levelwise constructions in the simplicial variable, so they hold degreewise. For (b), $R[z]$ is the part of $R[z^{\pm 1}]$ of nonnegative degree, so $\psi(A_w) \subset R[z]$ fails exactly when $aw < 0$ and $x_w \neq 0$. The elements with $aw < 0$ generate I_a, and a ring map killing a generating set of an ideal kills the ideal. $\square$

Nothing in Lemma A.3 required A to be resolved: the target $R[z]$ is a polynomial extension of R, and the graded pieces were used only to split a map of sets. The next two statements record the consequences that the proof of Theorem 8.9 uses.

Lemma A.4 (Sign dependence)**.** *For $a \neq 0$ the ideal I_a of* (77) *depends only on the sign of a. Consequently, for two exponents a, a' of the same sign, the attractor algebras $A^a = A^{a'}$, the induced strata, and the restriction maps to the open orbit coincide; only the universal section $z \mapsto z^a x$ of* (76) *differs. For $a = 0$ one has $I_0 = 0$ and $A^0 = A$.*

Proof. The condition $aw < 0$ is the condition $\operatorname{sign}(a)w < 0$, which depends on a only through its sign; for $a = 0$ it is empty. The strata and the restriction maps are defined from A^a and the inclusion $\operatorname{Spec} A^a \subset \operatorname{Spec} A$, so they inherit this. The section (76) is $z \mapsto z^a x$, which changes with a. $\square$

The value $a = 0$ is the degenerate case in which the extension condition is vacuous, the limit point need not be fixed, and no fixed component F is selected. This is why exponent zero occurs in Theorem 8.9 as a removed threshold rather than as an interior point of a tail.

Proposition A.5 (One-chart equivalence)**.** *Let $a \neq 0$ and let F be a connected component of $W^{\mathbf{G}_m}$. Write $\mathfrak{S}_a^0$ for the stack of rotation-fixed sections over the chart P^0 with exponent a, imposing membership in W^{gen} and limit in F. Then evaluation at $z = 1$ is an equivalence of derived stacks*

$$\mathrm{ev}_1 \colon \mathfrak{S}_a^0 \xrightarrow{\sim} [W_F^a/\mathbf{G}_m],$$

natural in the simplicial test algebra, compatible with arbitrary derived base change, and independent of the chosen invariant affine chart. In particular $\mathfrak{S}_a^0$ has no derived structure, and its classical truncation is the Białynicki–Birula stratum W_F^a modulo the gauge group, including stabilizers.

Proof. Work first on $\operatorname{Spec} A$ and before taking the quotient by the gauge group. By Lemma A.3 the sections over P^0 with exponent a and values in $\operatorname{Spec} A$ are, via evaluation at 1, the R-points of $\operatorname{Spec} A^a$; this is an isomorphism of functors on simplicial commutative $\mathbf{C}$-algebras, hence commutes with base change, since (76) is levelwise and does not involve R. For $a > 0$ the ideal I_a is generated by the negative weight spaces, so $\operatorname{Spec} A^a$ is the locus where $\lim_{t\to 0} t \cdot x$ exists; for $a < 0$ it is the locus where the opposite limit exists. The subalgebra of positive-degree elements of A^a is an ideal, so the projection to the weight-zero part is a ring map and defines the limit morphism $\operatorname{Spec} A^a \to \operatorname{Spec} A_0$ onto the fixed locus of the chart. The preimage of a connected component F is open and closed in $\operatorname{Spec} A^a$; intersecting with the open W^{gen} imposes generic semistability. This is the chartwise description of W_F^a.

Chart independence. Because W is projective, the limit $\lim_{t\to 0} t^a x$ exists in W for every x, and because W is separated an extension of a section across the origin is unique when it exists. Two invariant affine charts therefore impose the same condition on their overlap, and the chartwise closed subschemes glue to the locally closed subscheme $W_F^a \subset W$. Uniqueness also makes the identification ev_1 compatible on overlaps, so it globalizes.

Descent to the quotient stack. A rotation-fixed section over P_R^0 with values in $[W/\mathbf{G}_m]$ is, locally on $\operatorname{Spec} R$, a trivialization of the $\mathbf{G}_m$-bundle together with a section as above; two trivializations differ by a rotation-invariant gauge transformation, and the rotation-invariant gauge transformations over P^0 are the constants $\mathbf{G}_m(R)$. Quotienting by this action turns the previous paragraph into the asserted equivalence with $[W_F^a/\mathbf{G}_m]$, stabilizers included.

Absence of derived structure. The chart with its rotation action is the quotient stack $[\mathbf{A}^1/\mathbf{C}^\times_{\mathrm{rot}}]$. Its quasicoherent cohomology vanishes in positive degrees, since cohomology on $\mathbf{A}^1$ vanishes there and taking rotation invariants is exact in characteristic zero. Hence the deformation theory of $\mathfrak{S}_a^0$ relative to a point is concentrated in degree zero, so the stack is smooth and its structure sheaf is discrete. This is consistent with Białynicki–Birula: for smooth W the fixed components and their attracting loci are smooth. □

Proposition A.6 (Two-chart descent)**.** *With $\mathfrak{S}_{a_-,a_+}$ as in Definition A.1, evaluation at 1 is an equivalence of derived stacks*

$$\mathrm{ev}_1 \colon \mathfrak{S}_{a_-,a_+} \xrightarrow{\sim} \mathfrak{Z} = [W_{F_-}^{a_-}/\mathbf{G}_m] \underset{[W/\mathbf{G}_m]}{\overset{\mathbf{R}}{\times}} [W_{F_+}^{a_+}/\mathbf{G}_m],$$

and the inverse equivalence equips a family x with the two prescribed clutching bundles and the sections $z \mapsto z^{a_\pm} x$. By Lemma A.4, the right-hand side depends on the pair (a_-, a_+) only through the two signs and the two components F_-, F_+.

Proof. The two charts cover P and meet in the open orbit. A section over P_R is a pair of sections over the two charts together with an identification of their restrictions to the open orbit, and mapping into a stack makes this a homotopy limit rather than a strict equality of restrictions; that homotopy limit is the homotopy fibre product of the two chart-section stacks over the section stack of the open orbit. The rotation-fixed sections over the open orbit are, by evaluation at 1 and the argument of Proposition A.5 with $I_a = 0$, the stack $[W/\mathbf{G}_m]$; the restriction maps from the two charts are the two stratum inclusions. Applying Proposition A.5 to each chart gives the displayed equivalence. Since evaluation at 1 is the common value of the two extensions, the inverse construction reads off x and reconstructs the two sections by (76), which is where the exponents reappear. □

A.3 Tangent complexes and the 2-commutative square

All complexes in this subsection are relative to the fixed moduli $\mathfrak{M}$ of marked domains. Let $\pi\colon \mathcal{P} \to \mathfrak{Z}$ be the universal domain, $u_a\colon \mathcal{P} \to [W/\mathbf{G}_m]$ the universal gauged map with exponents $a = (a_-, a_+)$, and

$$V := \mathrm{ev}_1^* T([W/\mathbf{G}_m])$$

the pullback of the tangent complex of the quotient stack along evaluation at 1. Define $F^0_{\leq 0}V$ and $F^\infty_{\leq 0}V$ to be the subcomplexes of V consisting of the rotation-invariant sections of $u_a^* T([W/\mathbf{G}_m])$ over the chart at 0, respectively at ∞, viewed inside V by restriction to the open orbit. These are the extension subcomplexes of the in-text display; the subscript records which half of the weight decomposition at the corresponding limit point they select, and only the intrinsic description just given is used below. No equivariant splitting of V is chosen.

Lemma A.7 (Invariant Čech presentation)**.** *There are canonical equivalences, natural in* $\mathfrak{Z}$*,*

$$\left(R\pi_* u_a^* T([W/\mathbf{G}_m])\right)^{\mathrm{rot}} \simeq \left[F^0_{\leq 0}V \oplus F^\infty_{\leq 0}V \longrightarrow V\right] \simeq T_{\mathfrak{Z}/\mathfrak{M}}, \tag{78}$$

where the middle complex sits in degrees 0 *and* 1 *and its differential is the difference of the two restrictions to the open orbit. Moreover* $F^0_{\leq 0}V$ *and* $F^\infty_{\leq 0}V$ *are the pullbacks of* $T([W^{a_-}_{F_-}/\mathbf{G}_m])$ *and* $T([W^{a_+}_{F_+}/\mathbf{G}_m])$*, and all three terms depend on* a *only through the two signs.*

Proof. The pullback $u_a^* T([W/\mathbf{G}_m])$ is a two-term complex of vector bundles on $\mathcal{P}$, and the two charts are affine over $\mathfrak{Z}$, so the Čech complex of the two-chart cover computes $R\pi_*$. It is $[\,\Gamma(P^0) \oplus \Gamma(P^\infty) \to \Gamma(\mathbf{G}_m)\,]$ with the difference differential. Taking $\mathbf{C}^\times_{\mathrm{rot}}$-invariants is exact in characteristic zero, because the rotation group is linearly reductive and invariants are the weight-zero summand of a weight decomposition; so invariants may be taken termwise. By definition the invariants of the three terms are $F^0_{\leq 0}V$, $F^\infty_{\leq 0}V$ and V, the last because a rotation-invariant section over the open orbit is determined by its value at $z = 1$. This gives the first equivalence.

For the identification of the two extension subcomplexes, apply Lemma A.3 to the module of sections rather than to the ring: on an invariant affine chart, a graded A-module $M = \bigoplus_w M_w$ gives $(M \otimes_A R[z])^{\mathrm{rot}} = \bigoplus_{aw \geq 0}(M_w \otimes_A R)$ and $(M \otimes_A R[z^{\pm 1}])^{\mathrm{rot}} = \bigoplus_w (M_w \otimes_A R)$. Taking M to be the module of sections of $T([W/\mathbf{G}_m])$ identifies the invariants over the chart with the tangent complex of the attractor subscheme, which is the content of Białynicki–Birula's description of the attracting locus of a smooth $\mathbf{G}_m$-variety. The same statement in family form over a simplicial base is the module version of Proposition A.5.

The second equivalence in (78) is the tangent complex of a homotopy fibre product: for $\mathfrak{Z}$ as in Proposition A.6 there is a fibre sequence $T_{\mathfrak{Z}/\mathfrak{M}} \to T_- \oplus T_+ \to V$ with $T_\pm = T([W^{a\pm}_{F_\pm}/\mathbf{G}_m])$ pulled back, which is the displayed two-term complex. Finally, sign dependence follows from Lemma A.4 applied to the module computation above. □

Remark A.8 (Where the degree-one term comes from). The complex (78) has h^1 equal to the cokernel of $F^0_{\leq 0}V \oplus F^{\infty}_{\leq 0}V \to V$, which is nonzero exactly when the two attracting loci fail to meet transversely at the point in question. The gauge direction contributes to all three terms and cancels in the difference; on the generic semistable locus the stabilizers are finite, so h^{-1} vanishes and the complex is supported in degrees 0 and 1. Along one tail the two signs are constant and the zero-weight changes were removed as thresholds, so the two subcomplexes and the restriction map are constant, while the universal section $z \mapsto z^{a_\pm}x$ is not.

Lemma A.9 (Woodward's obstruction theory as a truncation)**.** *Put*

$$E_a = \left((R\pi_* u_a^* T([W/\mathbf{G}_m]))^{\mathrm{rot}}\right)^\vee.$$

The cotangent complex of the derived mapping stack gives $E_a \simeq \tau_{[-1,0]}L_{\mathfrak{S}_{a_-,a_+}/\mathfrak{M}}$, and under this identification the morphism $E_a \to L_{t_0\mathfrak{S}_{a_-,a_+}/\mathfrak{M}}$ of [Woo18, Definition 7.13], in its relative form over the twisted domain stack [Woo18, Section 8.3], [Woo17, Example 6.6(c)], is the composite

$$\tau_{[-1,0]}L_{\mathfrak{S}_{a_-,a_+}/\mathfrak{M}} \longrightarrow L_{\mathfrak{S}_{a_-,a_+}/\mathfrak{M}} \longrightarrow L_{t_0\mathfrak{S}_{a_-,a_+}/\mathfrak{M}}.$$

Proof. The first assertion is the cotangent complex formula for a mapping stack [TV08]: the relative cotangent complex of $\mathfrak{S}_{a_-,a_+}$ over $\mathfrak{M}$ is the pushforward along π of the pullback of the cotangent complex of the target, dualized as above, and the rotation-invariant part appears because the sections are rotation-fixed. Both sides are perfect of amplitude $[-1,0]$ by Lemma A.7.

For the second assertion, both morphisms are the map classifying the deformation theory of the same universal object: Woodward's is obtained from the Kodaira–Spencer map of the universal gauged map relative to the domain stack, which is exactly the map that the derived structure of the mapping stack induces on the classical truncation. In the presentation of Lemma A.7 both are represented by the same map, namely the dual of the map that sends a first-order deformation of a section to the induced pair of deformations over the two charts. We use here the standard comparison between a classical obstruction theory built from the deformation theory of a universal object and the truncation of the derived cotangent complex; that comparison is imported from [STV15], not reproved. $\square$

Proposition A.10 (The 2-commutative square)**.** *Under the equivalence* $\mathrm{ev}_1\colon \mathfrak{S}_{a_-,a_+} \simeq \mathfrak{Z}$ *of Proposition A.6, the square*

$$\begin{array}{ccc} E_a & \longrightarrow & L_{t_0\mathfrak{S}_{a_-,a_+}/\mathfrak{M}} \\ \downarrow\sim & & \downarrow\sim \\ \tau_{[-1,0]}L_{\mathfrak{Z}/\mathfrak{M}} & \longrightarrow & L_{t_0\mathfrak{Z}/\mathfrak{M}} \end{array}$$

is 2-commutative, its left vertical arrow is a quasi-isomorphism, and the two perfect obstruction theories are isomorphic as morphisms to the cotangent complex. Passing to classical truncations, the derived data recover Woodward's relative perfect obstruction theory, hence his relative virtual class.

Proof. By Lemma A.9 the upper horizontal map is the truncation map of $L_{\mathfrak{S}_{a_-,a_+}/\mathfrak{M}}$; the lower horizontal map is by definition the truncation map of $L_{\mathfrak{Z}/\mathfrak{M}}$. The assignment $\mathfrak{X} \mapsto (\tau_{[-1,0]}L_{\mathfrak{X}/\mathfrak{M}} \to L_{t_0\mathfrak{X}/\mathfrak{M}})$ is functorial for equivalences of derived stacks over $\mathfrak{M}$, and the two vertical arrows are the values of that functor on ev_1. The square is therefore the naturality square of a natural transformation evaluated at an equivalence, and the required homotopy is the one supplied by ev_1.

It remains to check that this homotopy is the one asserted in the proof of Theorem 8.9, namely the differential of evaluation at 1 on the two-chart universal section. In the presentation (78) this is an equality of representatives rather than a homotopy. Indeed, both composites in the square are, after dualizing, computed on a first-order deformation as follows. Restrict the deformation of the section to the two charts; each restriction is a section of the extension subcomplex at that chart by Lemma A.7; then restrict both to the open orbit and take the difference in V. For the upper route this is the Čech differential of $R\pi_* u_a^* T([W/\mathbf{G}_m])$; for the lower route it is the structure map of the homotopy fibre product $\mathfrak{Z}$. Both are the map $F^0_{\leq 0}V \oplus F^\infty_{\leq 0}V \to V$ given by the difference of the two restrictions to the open orbit, and these restrictions are the same morphisms of sheaves in the two cases, because the universal section restricted to a chart is the composite of the stratum inclusion with the chart section (76), and the latter is an isomorphism onto its image after evaluation at 1. The two representatives are equal, so the comparison homotopy may be taken to be the identity.

Since ev_1 is an equivalence, its induced map on $\tau_{[-1,0]}L_{-/\mathfrak{M}}$ is a quasi-isomorphism; this is the left vertical arrow. A commuting square whose verticals are quasi-isomorphisms is an isomorphism of the two morphisms to the cotangent complex, which is the assertion for obstruction theories. Taking t_0 and using Lemma A.9 identifies the upper row with Woodward's relative perfect obstruction theory, and an isomorphism of relative perfect obstruction theories over a common base induces equality of the associated relative virtual classes in Vistoli's bivariant Chow groups [Woo17, Example 6.6(c)]. □

A.4 Orbifold charts, μ_k-equivariance, and rigidification

Woodward's clutching construction has an orbifold form [Woo18, Definition 9.7]: one passes to a k-fold cover of the open orbit, the cocharacters $\widetilde{\varphi}_\pm$ are taken with values in $\frac{1}{k}\mathbf{Z}$, and the section on a chart is $z \mapsto \widetilde{\varphi}_\pm(z^{1/k})x$. The chart is then $[\operatorname{Spec} R[z^{1/k}]/\mu_k]$ with μ_k acting by $z^{1/k} \mapsto \theta z^{1/k}$ for a primitive k-th root of unity θ.

Proposition A.11 (Equivariance and descent)**.** *Write $a = b/k$ with $b \in \mathbf{Z}$. Then:*

(a) *Lemma A.3 holds verbatim with $R[z]$ replaced by $R[z^{1/k}]$ and $aw \in \frac{1}{k}\mathbf{Z}$; the attractor ideal is unchanged, being $(A_w : bw < 0)$.*

(b) *The stabilizer element of the clutching datum is $\widetilde{\varphi}(\theta) = \theta^{\,b}$, so the inertia label, the μ_k-weights of the universal bundle at the orbifold point, and the twisted-sector evaluation depend on b only through its class modulo k. Within one tail, consecutive affine degrees differ by k and the sign of b is constant.*

(c) *Taking μ_k-invariants is exact in characteristic zero, and the equivalence of Proposition A.6 and the square of Proposition A.10 are μ_k-equivariant.*

(d) *Rigidification replaces the inertia stack by its quotient by the finite central $B\mu_k$, $\bar{I}_{X,r} = I_{X,r}/B\mu_r$ [Woo17, Section 4.3, Proposition 4.3(f),(g)]. Pullback along this gerbe is fully faithful with $\pi_*\pi^* \simeq \mathrm{id}$, hence conservative, so the equivalence and the obstruction theory descend. Stabilizer and graph-automorphism groups, and their localization multiplicities, are unchanged.*

Proof. (a) The proof of Lemma A.3 used only that the target is a graded polynomial extension of R whose degrees are bounded below by zero. With $z^{1/k}$ of degree $1/k$ the same splitting applies, and $aw < 0$ is the condition $bw < 0$.

(b) The stabilizer of the orbifold point acts through $\widetilde{\varphi}(\theta)$, which for $G = \mathbf{G}_m$ and $\widetilde{\varphi}(t) = t^{b/k}$ is $\theta^{\,b}$; this depends only on $b \bmod k$. The isotropy representation of the universal bundle at the orbifold point is determined by the same element, and the twisted-sector evaluation of [Woo18, Corollary 9.10] is the corresponding component of the inertia stack. Fixing the stabilizer congruence class of Theorem 8.9 is exactly fixing $b \bmod k$, so a tail consists of the b in one class with one sign, and its consecutive elements differ by k.

(c) Exactness of invariants under a finite group in characteristic zero is standard, and it is the only property of μ_k used in Lemma A.7. All the constructions of Sections A.2 and A.3 are defined by the graded formula (76) and by restriction morphisms, both of which commute with the μ_k-action on $z^{1/k}$; the equivalence and the square are therefore diagrams of μ_k-equivariant objects and morphisms.

(d) A $B\mu_k$-gerbe in characteristic zero has $\pi_*\pi^* \simeq \mathrm{id}$ on quasicoherent complexes, because π_* takes the weight-zero part of the μ_k-decomposition and a pulled-back complex is concentrated in weight zero. Hence π^* is fully faithful and conservative, and a morphism of pulled-back objects which is μ_k-equivariant descends and is an equivalence downstairs precisely when it is one upstairs. The equivalence of Proposition A.6 and the square of Proposition A.10 are of that form by (c). The fibre products of [Woo18, equations (54) and (57)] are taken over the unrigidified inertia stack, so the constructions above take place there; passage to the rigidified inertia stack used for the evaluation maps changes rational cohomology only by the factors of r on r-twisted sectors recorded in [Woo17, Proposition 4.3(f),(g)], and those factors depend only on the order of the stabilizer, which is constant along a tail by (b). The parametrized principal component has no domain automorphisms; the fixed bubble factors retain theirs, and their automorphism groups are the ones appearing in the localization multiplicities. □

A.5 Attached bubbles, cutting, and reduction between Artin levels

Proposition A.12 (Attached factors and cutting)**.** *Let $\mathfrak{F}$ be the fixed graph stack obtained from $\mathfrak{S}_{a_-,a_+}$ by attaching the fixed stable-map factors of [Woo18, equations (54)–(57)] along the two inertia evaluations, as a derived fibre product. Then:*

(a) *the relative cotangent complex of $\mathfrak{F}$ fits in the exact triangle whose outer terms are the direct sum of the factor obstruction theories and the pullback of the cotangent complex of the inertia stack;*

(b) *the identification of Proposition A.10 is the identity on the stable-map and inertia terms, hence induces an isomorphism of these triangles;*

(c) *consequently the identification is compatible with the cutting axiom [Woo18, Proposition 7.14(b)], whose proof identifies the gauged moduli stack with a fibre product over the diagonal [Woo17, Proposition 5.21], and it identifies the resulting virtual classes;*

(d) *every construction above is a derived fibre product along finitely many morphisms of stacks over $\mathbf{C}$; the coefficient ring enters only through the localization coefficients, so the identifications commute with the reduction maps between finite Artin levels.*

For the irreducible unstable convention there is no stable-map factor: by [Woo18, Example 9.15] the domain is irreducible and the normal complex is the moving index alone, so the node and attaching factors of (52) *are absent and the square of Proposition A.10 applies with no gluing term.*

Proof. (a) is the cotangent triangle of a derived fibre product, applied to $\mathfrak{F} = \mathfrak{F}_- \times^{\mathbf{R}}_I \mathfrak{F}_+$ with I the inertia stack of the quotient: the triangle reads $\mathrm{ev}^* L_I \to L_{\mathfrak{F}_-} \oplus L_{\mathfrak{F}_+} \to L_{\mathfrak{F}}$ after pullback to $\mathfrak{F}$. This is the derived form of the exact triangle by which Woodward derives cutting-edge compatibility for the index complexes [Woo18, Equation (35) and Section 7.1].

(b) The degree shift changes only the principal factor. The attached stable-map factors and their obstruction theories are the fixed proper stable-map data at a fixed ordinary degree and marking set, and the inertia term is determined by the stabilizer congruence class, which is fixed along a tail by Proposition A.11(b). Hence the comparison map is the identity on those two terms, and Proposition A.10 supplies the map on the third; a map of triangles with two identity legs and one quasi-isomorphism is an isomorphism of triangles.

(c) Woodward's cutting axiom expresses the virtual class of the glued type as the Gysin pullback $\Delta^!$ of the virtual class of the cut type, using the identification of the moduli stack with a fibre product over the diagonal. The Gysin map is determined by the obstruction theory of the factors and the regular embedding Δ; by (a) and (b) both are identified, so the two cut classes agree.

(d) Each construction is one of: the equivalence of Proposition A.6, a homotopy fibre product along evaluation maps, or the truncation map of a cotangent complex. All are morphisms of stacks and of complexes over $\mathbf{C}$, independent of the Novikov and bulk coefficients; a finite Artin quotient enters only when the resulting virtual classes are paired with localization coefficients. The identifications are therefore compatible with every reduction $R_{N+1} \to R_N$. Compatibility here is a statement about the identifications, not about the numbers they produce: the identified stacks, obstruction theories, and virtual classes carry no N, so there is nothing for a reduction to act on, and the assertion is automatic rather than a level-by-level verification. The paired localization coefficients do depend on N, and for them compatibility means the usual thing, that reduction sends the level-$N+1$ pairing to the level-N one. □

Remark A.13 (Imported statements). Two comparisons are used above without proof. The first, from [TV08], is the cotangent complex formula for a derived mapping stack, used in Lemma A.9 to identify E_a with $\tau_{[-1,0]} L_{\mathfrak{S}_{a_-,a_+}/\mathfrak{M}}$. The second, from [STV15], is the identification of a classical perfect obstruction theory built from the deformation theory of a universal object with the classical truncation of the derived one; it is used in the same lemma to recognize [Woo18, Definition 7.13] in that truncation. Everything else is either the graded computation of Lemma A.3 and its module form, or a statement of Woodward's cited at the locators given at the start of this appendix. In particular, no derived statement is used at any point where a virtual class or a localization contribution is finally evaluated: those are read off from the classical truncations, in the form in which [Woo18] states them.

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